

Cancer Statistics, 2014

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Each year, the American Cancer Society estimates the numbers of new cancer cases and deaths that will occur in the United States in the current year and compiles the most recent data on cancer incidence, mortality, and survival. Incidence data were collected by the National Cancer Institute, the Centers for Disease Control and Prevention, and the North American Association of Central Cancer Registries and mortality data were collected by the National Center for Health Statistics. A total of 1,665,540 new cancer cases and 585,720 cancer deaths are projected to occur in the United States in 2014. During the most recent 5 years for which there are data (2006-2010), delay-adjusted cancer incidence rates declined slightly in men (by 0.6% per year) and were stable in women, while cancer death rates decreased by 1.8% per year in men and by 1.4% per year in women. The combined cancer death rate (deaths per 100,000 population) has been continuously declining for 2 decades, from a peak of 215.1 in 1991 to 171.8 in 2010. This 20% decline translates to the avoidance of approximately 1,340,400 cancer deaths (952,700 among men and 387,700 among women) during this time period. The magnitude of the decline in cancer death rates from 1991 to 2010 varies substantially by age, race, and sex, ranging from no decline among white women aged 80 years and older to a 55% decline among black men aged 40 years to 49 years. Notably, black men experienced the largest drop within every 10-year age group. Further progress can be accelerated by applying existing cancer control knowledge across all segments of the population. *CA Cancer J Clin* 2014;64:9-29. © 2014 American Cancer Society, Inc.

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Introduction

Cancer is a major public health problem in the United States and many other parts of the world. One in 4 deaths in the United States is due to cancer. In this article, we provide the expected numbers of new cancer cases and deaths in 2014 in the United States nationally and for each state, as well as a comprehensive overview of cancer incidence, mortality, and survival rates and trends using the most current population-based data available. In addition, we estimate the total number of deaths averted since the early 1990s as a result of 2 decades of declining cancer death rates and present the actual number of deaths reported in 2010 by age for the 10 leading causes of death and the 5 leading causes of cancer death.

Materials and Methods

Incidence and Mortality Data

Mortality data from 1930 to 2010 were obtained from the National Center for Health Statistics (NCHS).¹ Population-based cancer incidence data in the United States are collected both by the National Cancer Institute's (NCI's) Surveillance, Epidemiology, and End Results (SEER) Program and the Centers for Disease Control and Prevention's (CDC's) National Program of Cancer Registries (NPCR). The SEER program reports long-term (beginning in 1973), high-quality incidence, prevalence, and survival data. Long-term incidence and survival trends (1975-2010) were based on data from the 9 oldest SEER areas (Connecticut, Iowa, Hawaii, New Mexico, Utah, and the metropolitan areas of Atlanta, Detroit, San Francisco-Oakland, and Seattle-Puget Sound), representing approximately 10% of the US population.² As of 1992, SEER data have been available for 4 additional populations (Alaska Natives, Los Angeles County, San Jose-Monterey, and rural Georgia) that increase the coverage of minority groups and allow for stratification by race and ethnicity.³ Data from these

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TABLE 1. Estimated New Cancer Cases and Deaths by Sex, United States, 2014*

	ESTIMATED NEW CASES			ESTIMATED DEATHS		
	BOTH SEXES	MALE	FEMALE	BOTH SEXES	MALE	FEMALE
All sites	1,665,540	855,220	810,320	585,720	310,010	275,710
Oral cavity & pharynx	42,440	30,220	12,220	8,390	5,730	2,660
Tongue	13,590	9,720	3,870	2,150	1,450	700
Mouth	11,920	7,150	4,770	2,070	1,130	940
Pharynx	14,410	11,550	2,860	2,540	1,900	640
Other oral cavity	2,520	1,800	720	1,630	1,250	380
Digestive system	289,610	162,730	126,880	147,260	84,970	62,290
Esophagus	18,170	14,660	3,510	15,450	12,450	3,000
Stomach	22,220	13,730	8,490	10,990	6,720	4,270
Small intestine	9,160	4,880	4,280	1,210	640	570
Colon†	96,830	48,450	48,380	50,310	26,270	24,040
Rectum	40,000	23,380	16,620			
Anus, anal canal, & anorectum	7,210	2,660	4,550	950	370	580
Liver & intrahepatic bile duct	33,190	24,600	8,590	23,000	15,870	7,130
Gallbladder & other biliary	10,650	4,960	5,690	3,630	1,610	2,020
Pancreas	46,420	23,530	22,890	39,590	20,170	19,420
Other digestive organs	5,760	1,880	3,880	2,130	870	1,260
Respiratory system	242,550	130,000	112,550	163,660	90,280	73,380
Larynx	12,630	10,000	2,630	3,610	2,870	740
Lung & bronchus	224,210	116,000	108,210	159,260	86,930	72,330
Other respiratory organs	5,710	4,000	1,710	790	480	310
Bones & joints	3,020	1,680	1,340	1,460	830	630
Soft tissue (including heart)	12,020	6,550	5,470	4,740	2,550	2,190
Skin (excluding basal & squamous)	81,220	46,630	34,590	12,980	8,840	4,140
Melanoma-skin	76,100	43,890	32,210	9,710	6,470	3,240
Other nonepithelial skin	5,120	2,740	2,380	3,270	2,370	900
Breast	235,030	2,360	232,670	40,430	430	40,000
Genital system	338,450	243,460	94,990	58,970	30,180	28,790
Uterine cervix	12,360		12,360	4,020		4,020
Uterine corpus	52,630		52,630	8,590		8,590
Ovary	21,980		21,980	14,270		14,270
Vulva	4,850		4,850	1,030		1,030
Vagina & other genital, female	3,170		3,170	880		880
Prostate	233,000	233,000		29,480	29,480	
Testis	8,820	8,820		380	380	
Penis & other genital, male	1,640	1,640		320	320	
Urinary system	141,610	97,420	44,190	30,350	20,610	9,740
Urinary bladder	74,690	56,390	18,300	15,580	11,170	4,410
Kidney & renal pelvis	63,920	39,140	24,780	13,860	8,900	4,960
Ureter & other urinary organs	3,000	1,890	1,110	910	540	370
Eye & orbit	2,730	1,440	1,290	310	130	180
Brain & other nervous system	23,380	12,820	10,560	14,320	8,090	6,230
Endocrine system	65,630	16,600	49,030	2,820	1,300	1,520
Thyroid	62,980	15,190	47,790	1,890	830	1,060
Other endocrine	2,650	1,410	1,240	930	470	460
Lymphoma	79,990	43,340	36,650	20,170	11,140	9,030
Hodgkin lymphoma	9,190	5,070	4,120	1,180	670	510
Non-Hodgkin lymphoma	70,800	38,270	32,530	18,990	10,470	8,520
Myeloma	24,050	13,500	10,550	11,090	6,110	4,980
Leukemia	52,380	30,100	22,280	24,090	14,040	10,050
Acute lymphocytic leukemia	6,020	3,140	2,880	1,440	810	630
Chronic lymphocytic leukemia	15,720	9,100	6,620	4,600	2,800	1,800
Acute myeloid leukemia	18,860	11,530	7,330	10,460	6,010	4,450
Chronic myeloid leukemia	5,980	3,130	2,850	810	550	260
Other leukemia†	5,800	3,200	2,600	6,780	3,870	2,910
Other & unspecified primary sites‡	31,430	16,370	15,060	44,680	24,780	19,900

*Rounded to the nearest 10; estimated new cases exclude basal cell and squamous cell skin cancers and in situ carcinoma except urinary bladder.

About 62,570 carcinoma in situ of the female breast and 63,770 melanoma in situ will be newly diagnosed in 2014.

†Estimated deaths for colon and rectum cancers are combined.

‡More deaths than cases may reflect lack of specificity in recording underlying cause of death on death certificates and/or an undercount in the case estimate.

TABLE 2. Incidence Rates for All Cancers Combined (2006-2010) and Estimated New Cases* for Selected Cancers (2014) by State

STATE	INCIDENCE RATE†	ALL CASES	FEMALE BREAST	UTERINE CERVIX	COLON & RECTUM	UTERINE CORPUS	LEUKEMIA	LUNG & BRONCHUS	MELANOMA OF THE SKIN	NON-HODGKIN LYMPHOMA	PROSTATE	URINARY BLADDER
Alabama	469.6	26,770	3,660	210	2,350	650	690	4,160	1,320	990	3,760	990
Alaska	469.3	3,750	450	‡	280	100	100	430	90	140	530	150
Arizona	401.1	32,830	4,520	210	2,560	910	950	4,280	1,430	1,320	4,390	1,490
Arkansas§	460.6	16,520	2,050	140	1,500	400	480	2,660	490	660	2,240	640
California	442.2	171,730	26,130	1,550	13,930	5,650	5,650	18,780	8,440	7,770	23,010	7,210
Colorado	432.5	23,810	3,780	160	1,720	750	870	2,540	1,400	1,060	3,680	1,040
Connecticut	505.7	22,070	3,160	120	1,650	790	610	2,730	1,090	920	3,120	1,170
Delaware	511.7	5,320	760	‡	420	180	150	790	290	220	800	260
Dist. of Columbia	486.9	2,840	430	‡	250	100	60	320	80	100	510	80
Florida	452.0	114,560	15,480	960	10,230	3,410	3,810	17,960	5,320	5,050	16,590	5,800
Georgia	471.3	47,390	7,010	420	3,940	1,310	1,370	6,540	2,180	1,820	7,600	1,710
Hawaii	430.1	6,640	1,090	60	700	270	220	890	410	300	810	250
Idaho	455.6	7,990	1,100	50	610	230	290	960	450	360	1,320	390
Illinois	491.2	66,840	9,230	470	5,530	2,290	2,180	9,100	2,440	2,890	8,820	3,090
Indiana	464.0	35,560	4,590	260	3,020	1,070	1,060	5,540	1,550	1,480	4,390	1,600
Iowa	485.6	17,630	2,320	100	1,580	610	640	2,330	980	800	2,340	830
Kansas	476.8	14,630	2,090	100	1,120	470	490	1,900	780	650	1,980	620
Kentucky	523.2	25,770	3,370	200	2,170	720	790	4,690	1,540	1,070	3,280	1,100
Louisiana	494.1	24,300	3,160	200	2,270	540	720	3,470	750	960	3,720	940
Maine	511.1	9,270	1,220	50	700	340	310	1,400	440	380	1,160	540
Maryland	462.0	30,680	4,570	230	2,500	1,020	800	3,990	1,400	1,210	5,000	1,280
Massachusetts	503.3	37,940	5,560	200	2,800	1,320	1,140	4,930	1,800	1,600	5,600	2,030
Michigan	492.3	58,610	7,660	340	4,570	2,010	1,830	8,090	2,830	2,500	8,740	2,930
Minnesota††	-	29,340	3,820	130	2,240	950	1,050	3,070	1,030	1,240	3,870	1,220
Mississippi	480.9	15,740	2,130	140	1,510	360	410	2,420	560	560	2,210	540
Missouri	467.7	33,890	4,610	240	2,970	1,090	1,040	5,370	1,510	1,430	4,010	1,530
Montana	466.3	5,850	860	‡	500	180	200	760	290	260	1,010	300
Nebraska	464.5	9,550	1,360	60	880	320	330	1,220	460	440	1,250	430
Nevada§	448.7	14,450	1,880	120	1,290	350	440	2,040	470	550	1,890	680
New Hampshire	507.2	8,450	1,150	‡	600	300	250	1,110	400	350	1,160	460
New Jersey	503.7	51,130	7,290	380	4,280	1,820	1,560	6,130	2,590	2,250	7,320	2,510
New Mexico	405.9	10,210	1,450	80	830	300	370	1,060	470	400	1,400	400
New York	504.1	107,200	15,230	850	8,590	4,040	3,460	13,720	4,240	4,720	15,440	5,330
North Carolina	477.2	52,550	7,580	380	4,230	1,570	1,550	7,850	2,540	2,110	7,580	2,170
North Dakota	460.0	3,730	510	‡	350	110	130	400	160	160	460	180
Ohio§	474.7	67,000	8,710	400	5,450	2,280	1,890	9,760	3,170	2,860	8,690	3,110
Oklahoma	475.8	19,830	2,700	160	1,760	530	660	3,320	650	850	2,570	830
Oregon	462.2	22,530	3,320	130	1,540	720	640	2,950	1,440	960	3,200	1,080
Pennsylvania	502.6	79,920	10,660	500	6,790	2,840	2,420	10,290	3,820	3,420	10,930	4,070
Rhode Island	506.5	6,370	870	‡	500	230	180	870	260	250	840	340
South Carolina	463.8	26,390	3,750	210	2,200	750	790	4,130	1,350	1,030	4,000	1,100
South Dakota	439.7	4,490	600	‡	410	150	160	540	200	200	590	210
Tennessee	475.8	36,570	4,840	290	3,030	930	1,040	5,980	1,910	1,470	4,670	1,510
Texas	441.5	115,730	16,080	1,140	9,760	3,130	4,190	14,890	3,420	5,030	15,900	4,190
Utah	418.6	10,780	1,440	60	650	350	390	650	770	490	1,780	420
Vermont	489.8	4,130	560	‡	290	140	110	550	220	160	580	210
Virginia	449.0	40,970	6,170	290	3,280	1,300	1,080	5,580	2,130	1,640	6,330	1,700
Washington	483.0	38,230	5,620	230	2,670	1,160	1,250	4,630	2,410	1,710	5,380	1,730
West Virginia	484.6	11,700	1,350	90	1,060	380	330	2,090	540	480	1,450	570
Wisconsin	466.0	32,480	4,330	190	2,520	1,140	1,150	4,020	1,440	1,410	4,630	1,580
Wyoming	435.0	2,890	420	‡	240	90	90	330	150	120	490	140
United States	469.6	1,665,540	232,670	12,360	136,830	52,630	52,380	224,210	76,100	70,800	233,000	74,690

*Rounded to the nearest 10; excludes basal cell and squamous cell skin cancers and in situ carcinomas except urinary bladder.

†Rates are per 100,000 and age adjusted to the 2000 US standard population. US rate excludes cases from Arkansas, Minnesota, Nevada, Ohio, and Virginia.

‡Estimate is fewer than 50 cases.

§High-quality incidence data were not available for all 5 years. Arkansas rate is based on cases diagnosed during 2006 to 2008; Nevada and Ohio rates are based on 2006 to 2009.

††Incidence data were not available.

Note: These model-based estimates are offered as a rough guide and should be interpreted with caution. State estimates may not add to US total due to rounding.

Estimated New Cases*

			Males	Females			
Prostate	233,000	27%			Breast	232,670	29%
Lung & bronchus	116,000	14%			Lung & bronchus	108,210	13%
Colorectum	71,830	8%			Colorectum	65,000	8%
Urinary bladder	56,390	7%			Uterine corpus	52,630	6%
Melanoma of the skin	43,890	5%			Thyroid	47,790	6%
Kidney & renal pelvis	39,140	5%			Non-Hodgkin lymphoma	32,530	4%
Non-Hodgkin lymphoma	38,270	4%			Melanoma of the skin	32,210	4%
Oral cavity & pharynx	30,220	4%			Kidney & renal pelvis	24,780	3%
Leukemia	30,100	4%			Pancreas	22,890	3%
Liver & intrahepatic bile duct	24,600	3%			Leukemia	22,280	3%
All Sites	855,220	100%			All Sites	810,320	100%

Estimated Deaths

			Males	Females			
Lung & bronchus	86,930	28%			Lung & bronchus	72,330	26%
Prostate	29,480	10%			Breast	40,000	15%
Colorectum	26,270	8%			Colorectum	24,040	9%
Pancreas	20,170	7%			Pancreas	19,420	7%
Liver & intrahepatic bile duct	15,870	5%			Ovary	14,270	5%
Leukemia	14,040	5%			Leukemia	10,050	4%
Esophagus	12,450	4%			Uterine corpus	8,590	3%
Urinary bladder	11,170	4%			Non-Hodgkin lymphoma	8,520	3%
Non-Hodgkin lymphoma	10,470	3%			Liver & intrahepatic bile duct	7,130	3%
Kidney & renal pelvis	8,900	3%			Brain & other nervous system	6,230	2%
All Sites	310,010	100%			All Sites	275,710	100%

FIGURE 1. Ten Leading Cancer Types for the Estimated New Cancer Cases and Deaths by Sex, United States, 2014.

*Estimates are rounded to the nearest 10 and exclude basal cell and squamous cell skin cancers and in situ carcinoma except urinary bladder.

SEER 13 registries were the source for the annual percent change in incidence from 1992 to 2010. The SEER program added 5 additional catchment areas beginning with cases diagnosed in 2000 (greater California, greater Georgia, Kentucky, Louisiana, and New Jersey), achieving 28% population coverage.⁴ Data from all 18 SEER areas were the source for cancer stage distribution, stage-specific survival, and the lifetime probability of developing cancer. Much of the statistical information presented herein was adapted from data previously published in the *SEER Cancer Statistics Review, 1975–2010*.⁵

The North American Association of Central Cancer Registries (NAACCR) compiles and reports incidence data from 1995 onward for cancer registries that participate in the SEER program or the NPCR. These data approach 100% coverage of the US population and were the source for incidence rates by state and race/ethnicity, as well as the projection of new cancer cases in 2014.⁶ Some of the data

presented herein were previously published in volumes 1 and 2 of *Cancer in North America: 2006–2010*.^{7,8}

All cancer cases were classified according to the *International Classification of Diseases for Oncology*.⁹ The lifetime probability of cancer was calculated using the NCI's DevCan software (version 6.7.0).¹⁰ All incidence and death rates were age-standardized to the 2000 US standard population and expressed per 100,000 population, as calculated by NCI's SEER*Stat software (version 8.1.2).¹¹

Cancer incidence rates in this report were adjusted for delays in reporting whenever possible. This adjustment, which is available only for SEER data, is based on historic patterns of case ascertainment and accounts for anticipated future corrections to registry data primarily due to a lag in case reporting. Delay adjustment has the largest effect on the most recent years of data for cancers that are frequently diagnosed in outpatient settings (eg, melanoma, leukemia, and prostate cancer) and provides a more accurate portrayal

TABLE 3. Death Rates for All Cancers Combined (2006-2010) and Estimated Deaths* for Selected Cancers (2014) by State

STATE	DEATH RATE†	ALL SITES	BRAIN & OTHER NERVOUS SYSTEM	FEMALE BREAST	COLON & RECTUM	LEUKEMIA	LIVER & INTRAHEPATIC BILE DUCT	LUNG & BRONCHUS	NON-HODGKIN LYMPHOMA	OVARY	PANCREAS	PROSTATE
Alabama	196.4	10,510	270	690	950	410	350	3,310	310	280	630	540
Alaska	182.5	990	‡	70	90	‡	‡	270	‡	‡	60	60
Arizona	155.2	11,400	310	780	990	500	470	2,840	390	310	790	640
Arkansas	198.0	6,730	150	420	620	270	210	2,200	200	140	400	310
California	160.3	57,950	1,590	4,270	5,150	2,510	3,140	12,590	2,000	1,560	4,150	3,380
Colorado	152.8	7,480	240	530	670	330	300	1,720	240	240	510	450
Connecticut	166.4	6,880	180	470	460	290	250	1,760	220	180	520	390
Delaware	187.7	1,980	‡	120	160	70	90	600	60	50	130	90
Dist. of Columbia	199.0	1,010	‡	80	100	‡	60	230	‡	‡	80	80
Florida	168.4	42,740	920	2,770	3,560	1,760	1,620	12,050	1,430	940	2,890	2,170
Georgia	179.2	16,320	380	1,220	1,480	620	590	4,690	460	430	1,040	800
Hawaii	143.7	2,450	‡	140	230	90	140	580	80	60	210	120
Idaho	162.8	2,730	80	180	210	130	80	670	90	60	210	180
Illinois	183.9	24,020	540	1,610	2,190	1,020	810	6,570	780	560	1,610	1,190
Indiana	192.6	13,370	310	860	1,090	550	380	4,140	440	310	840	580
Iowa	174.3	6,380	190	390	570	280	190	1,780	230	180	410	330
Kansas	174.7	5,460	150	370	480	260	170	1,560	200	140	370	250
Kentucky	209.5	10,130	210	590	850	370	280	3,570	300	200	570	390
Louisiana	201.3	9,040	200	640	840	330	400	2,650	260	190	600	400
Maine	187.7	3,300	90	190	250	130	110	970	100	60	200	160
Maryland	179.0	10,500	240	820	890	390	400	2,760	300	270	760	550
Massachusetts	177.6	12,810	310	790	990	510	550	3,500	400	320	920	630
Michigan	185.1	20,800	550	1,400	1,680	910	710	5,990	720	480	1,480	890
Minnesota	168.9	9,750	260	620	780	460	360	2,480	340	240	650	540
Mississippi	203.5	6,350	140	420	640	250	240	1,990	180	120	380	330
Missouri	189.2	12,870	310	910	1,090	540	450	3,950	390	250	860	550
Montana	166.3	2,000	50	130	170	90	50	520	70	60	130	130
Nebraska	170.7	3,480	100	210	340	140	100	900	130	80	240	200
Nevada	179.0	4,790	140	380	480	190	220	1,420	140	100	370	280
New Hampshire	177.3	2,670	70	170	200	100	80	750	80	60	190	130
New Jersey	175.0	16,350	350	1,290	1,510	630	600	3,970	510	440	1,220	760
New Mexico	155.5	3,600	90	260	350	140	170	790	110	90	240	220
New York	166.6	34,440	790	2,390	2,970	1,440	1,470	8,790	1,110	910	2,540	1,760
North Carolina	182.8	18,980	410	1,310	1,500	720	660	5,700	560	430	1,190	920
North Dakota	164.8	1,270	‡	90	130	60	‡	310	‡	‡	80	80
Ohio	192.4	25,260	600	1,720	2,140	990	790	7,370	810	570	1,730	1,200
Oklahoma	193.3	7,980	200	510	690	320	280	2,440	270	180	470	370
Oregon	176.3	7,940	230	510	660	320	340	2,090	280	220	550	440
Pennsylvania	184.5	28,670	610	1,940	2,490	1,200	980	7,600	1,010	730	1,990	1,370
Rhode Island	177.5	2,140	50	130	160	90	80	580	60	50	130	100
South Carolina	187.6	9,950	220	670	840	360	370	2,970	280	230	610	510
South Dakota	168.4	1,610	50	110	150	70	50	440	50	‡	110	90
Tennessee	199.1	14,280	350	910	1,220	540	500	4,630	440	290	820	630
Texas	169.7	37,830	950	2,700	3,430	1,530	2,080	9,600	1,230	900	2,440	1,660
Utah	131.3	2,870	110	270	250	150	100	460	120	80	240	210
Vermont	177.9	1,340	‡	80	100	50	60	390	‡	‡	90	70
Virginia	179.3	14,750	350	1,090	1,240	570	520	4,110	460	380	1,010	730
Washington	175.0	12,550	380	820	970	540	550	3,270	430	360	880	730
West Virginia	201.2	4,680	100	270	420	170	120	1,480	160	100	230	190
Wisconsin	175.6	11,360	310	710	860	550	390	3,000	400	300	800	630
Wyoming	167.8	990	‡	60	90	70	‡	250	‡	‡	80	40
United States	176.4	585,720	14,320	40,000	50,310	24,090	23,000	159,260	18,990	14,270	39,590	29,480

*Rounded to the nearest 10.

†Rates are per 100,000 and age adjusted to the 2000 US standard population.

‡Estimate is fewer than 50 deaths.

Note: State estimates may not add to US total due to rounding and the exclusion of states with fewer than 50 deaths.

of the cancer burden in the most recent time period.¹² For example, melanoma incidence rates adjusted for reporting delays are 14% higher than unadjusted rates in the most

recent data year. Delay-adjusted rates were obtained from the SEER Canques database (surveillance.cancer.gov/delay/canques.html [accessed August 6, 2013]).

TABLE 4. Probability (%) of Developing Invasive Cancers Within Selected Age Intervals by Sex, United States, 2008 to 2010*

		BIRTH TO 49	50 TO 59	60 TO 69	70 AND OLDER	BIRTH TO DEATH
All sites†	Male	3.5 (1 in 29)	6.8 (1 in 15)	15.4 (1 in 6)	36.9 (1 in 3)	43.9 (1 in 2)
	Female	5.4 (1 in 19)	6.0 (1 in 17)	10.1 (1 in 10)	26.7 (1 in 4)	38.0 (1 in 3)
Kidney & renal pelvis	Male	0.2 (1 in 480)	0.3 (1 in 289)	0.6 (1 in 154)	1.3 (1 in 75)	2.1 (1 in 49)
	Female	0.1 (1 in 753)	0.2 (1 in 586)	0.3 (1 in 317)	0.7 (1 in 134)	1.2 (1 in 83)
Breast	Female	1.9 (1 in 53)	2.3 (1 in 43)	3.5 (1 in 29)	6.7 (1 in 15)	12.3 (1 in 8)
	Male	0.3 (1 in 305)	0.7 (1 in 144)	1.3 (1 in 76)	4.0 (1 in 25)	5.0 (1 in 20)
Colorectum	Female	0.3 (1 in 334)	0.5 (1 in 189)	0.9 (1 in 109)	3.7 (1 in 27)	4.6 (1 in 22)
	Male	0.2 (1 in 421)	0.2 (1 in 614)	0.4 (1 in 279)	1.3 (1 in 76)	1.7 (1 in 60)
Leukemia	Female	0.2 (1 in 526)	0.1 (1 in 979)	0.2 (1 in 475)	0.8 (1 in 120)	1.2 (1 in 86)
	Male	0.2 (1 in 548)	0.7 (1 in 134)	2.1 (1 in 47)	6.7 (1 in 15)	7.6 (1 in 13)
Lung & bronchus	Female	0.2 (1 in 522)	0.6 (1 in 171)	1.6 (1 in 62)	4.9 (1 in 20)	6.3 (1 in 16)
	Male	0.4 (1 in 284)	0.4 (1 in 134)	0.8 (1 in 129)	2.1 (1 in 48)	2.9 (1 in 34)
Melanoma of the skin‡	Female	0.5 (1 in 206)	0.3 (1 in 313)	0.4 (1 in 243)	0.9 (1 in 113)	1.9 (1 in 53)
	Male	0.3 (1 in 357)	0.3 (1 in 338)	0.6 (1 in 171)	1.8 (1 in 56)	2.4 (1 in 42)
Non-Hodgkin lymphoma	Female	0.2 (1 in 537)	0.2 (1 in 475)	0.4 (1 in 233)	1.4 (1 in 71)	1.9 (1 in 52)
	Male	0.3 (1 in 298)	2.3 (1 in 43)	6.4 (1 in 16)	11.2 (1 in 9)	15.3 (1 in 7)
Prostate	Female	0.3 (1 in 348)	0.1 (1 in 812)	0.1 (1 in 824)	0.2 (1 in 619)	0.7 (1 in 154)
	Male	0.3 (1 in 370)	0.6 (1 in 171)	0.9 (1 in 111)	1.3 (1 in 78)	2.7 (1 in 37)

*For people free of cancer at beginning of age interval.

†All sites excludes basal cell and squamous cell skin cancers and in situ cancers except urinary bladder.

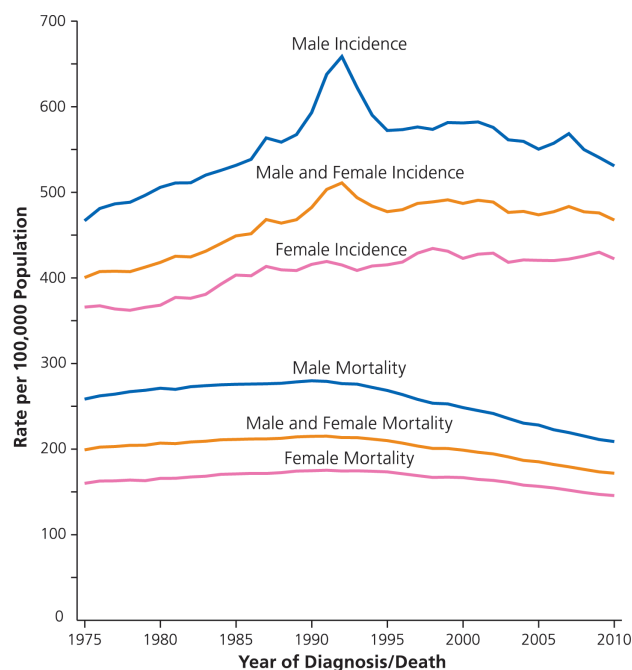
‡Probabilities for whites only.

Projected Cancer Cases and Deaths in 2014

The most recent year for which incidence and mortality data are available lags 3 to 4 years behind the current year due to the time required for data collection, compilation, quality control, and dissemination. Therefore, we project the numbers of new cancer cases and deaths in the United States in the current year in order to provide an estimate of the contemporary cancer burden. These estimates are not useful for tracking cancer occurrence over time because they are model-based and because the calculation methodology changes every few years in order to take advantage of improvements in modeling techniques, increased cancer registration coverage, and updated risk factor surveillance.

A 3-step spatio-temporal model was used to project the number of new invasive cancer cases that will be diagnosed in 2014 based on 1995 through 2010 high-quality incidence data from 49 states and the District of Columbia, representing 89% population coverage. (All states did not meet high quality data standards for all years and Minnesota did not submit incidence data to NAACCR during the 2012 call for data.) This method accounts for expected delays in case reporting and considers geographic variations in sociodemographic and lifestyle factors, medical settings, and cancer screening behaviors as predictors of incidence.¹³ First, complete incidence counts were estimated for each county from 1995 through 2010. Then these counts were adjusted to account for delays in cancer reporting. Finally, a temporal projection method (the vector autoregressive model) was applied to the last 15 years of data (1996–2010) to estimate counts for 2014,

which were then aggregated to obtain state-level estimates. This method cannot estimate numbers of basal cell or squamous cell skin cancers because data on the occurrence of these cancers are not reported to cancer registries. For the complete details of the case projection methodology, please refer to Zhu et al.¹⁴

**FIGURE 2. Trends in Cancer Incidence and Death Rates by Sex, United States, 1975 to 2010.**

Rates are age adjusted to the 2000 US standard population. Incidence rates are adjusted for delays in reporting.

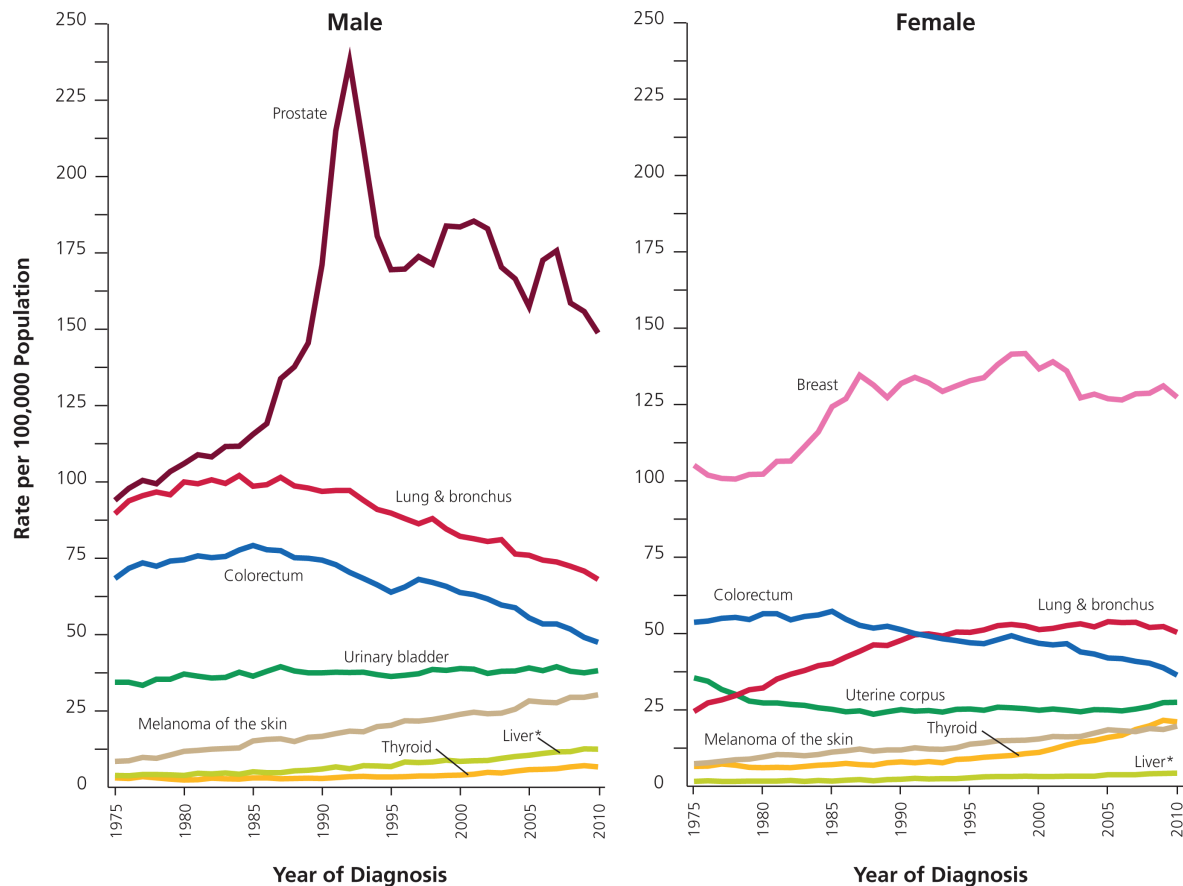


FIGURE 3. Trends in Incidence Rates for Selected Cancers by Sex, United States, 1975 to 2010.

Rates are age adjusted to the 2000 US standard population and adjusted for delays in reporting. *Includes intrahepatic bile duct.

To estimate the number of in situ female breast and melanoma cases diagnosed in 2014, we first estimated the number of cases occurring annually from 2001 through 2010 by applying age-specific SEER 13 incidence rates to the corresponding US population estimates provided in SEER*Stat.¹¹ (Delay-adjusted rates were available for in situ breast cancer but not for in situ melanoma.) We then projected the total number of cases in 2014 based on the average annual percent change in case counts from 2001 through 2010 generated by the joinpoint regression model.¹⁵

We estimated the number of cancer deaths expected to occur in 2014 in the United States overall and in each state using the joinpoint regression model based on the actual numbers of cancer deaths from 1995 through 2010 at the state and national levels as reported to the NCHS. For the complete details of this methodology, please refer to Chen et al.¹⁶

Other Statistics

The estimated number of cancer deaths averted in men and women due to the reduction in overall cancer death rates was calculated by first estimating the number of

cancer deaths that would have occurred if death rates had remained at their peak. The expected number of deaths was estimated by applying the 5-year age-specific cancer death rates in the peak year for age-standardized cancer death rates (1990 in men and 1991 in women) to the corresponding age-specific populations in the subsequent years through 2010. Then the difference between the number of expected and recorded cancer deaths in each age group and calendar year was summed for men and women separately.

Selected Findings

Expected Numbers of New Cancer Cases

Table 1 presents the estimated numbers of new cases of invasive cancer expected among men and women in the United States in 2014. The overall estimate of 1,665,540 new cases is the equivalent of more than 4,500 new cancer diagnoses each day. About 62,570 cases of breast carcinoma in situ and 63,770 cases of melanoma in situ are expected to be newly diagnosed in 2014. The estimated numbers of new cancer cases by state for selected cancer sites are shown in Table 2.

TABLE 5. Trends in Cancer Incidence (Delay-Adjusted) and Death Rates for Selected Cancers by Sex, United States, 1992 to 2010

	TREND 1		TREND 2		TREND 3		TREND 4		2006–2010 AAPC
	YEARS	APC	YEARS	APC	YEARS	APC	YEARS	APC	
All sites									
Incidence									
Overall	1992-1994	-3.2*	1994-1998	0.4	1998-2010	-0.4*			-0.4*
Male	1992-1994	-5.6*	1994-2010	-0.6*					-0.6*
Female	1992-1994	-0.4	1994-1998	1.2	1998-2003	-0.8*	2003-2010	0.1	0.1
Death									
Overall	1992-2001	-1.0*	2001-2010	-1.5*					-1.5*
Male	1992-2001	-1.4*	2001-2010	-1.8*					-1.8*
Female	1992-1995	-0.2	1995-1998	-1.2*	1998-2001	-0.4	2001-2010	-1.4*	-1.4*
Lung & bronchus									
Incidence									
Male	1992-2010	-1.9*							-1.9*
Female	1992-1998	0.8*	1998-2001	-1.3	2001-2005	0.7	2005-2010	-1.2*	-1.2*
Death									
Male	1992-2005	-1.9*	2005-2010	-2.9*					-2.9*
Female	1992-1996	1.1*	1996-2004	0.2	2004-2010	-1.4*			-1.4*
Colorectum									
Incidence									
Male	1992-1995	-2.6*	1995-1998	1.4	1998-2008	-2.5*	2008-2010	-4.2*	-3.3*
Female	1992-1995	-1.8*	1995-1998	1.8	1998-2008	-1.9*	2008-2010	-4.1*	-3.0*
Death									
Male	1992-2002	-2.0*	2002-2005	-4.0*	2005-2010	-2.5*			-2.5*
Female	1992-2001	-1.7*	2001-2010	-3.0*					-3.0*
Liver & intrahepatic bile duct									
Incidence									
Male	1992-2010	3.7*							3.7*
Female	1992-2010	2.9*							2.9*
Death									
Male	1992-2010	2.3*							2.3*
Female	1992-2010	1.4*							1.4*
Melanoma of skin									
Incidence									
Male	1992-2010	2.4*							2.4*
Female	1992-1997	3.9*	1997-2010	1.7*					1.7*
Death									
Male	1992-2010	0.4*							0.4*
Female	1992-2010	-0.5*							-0.5*
Thyroid									
Incidence									
Male	1992-1996	-0.9	1996-2010	5.4*					5.4*
Female	1992-1998	3.9*	1998-2010	6.5*					6.5*
Death									
Male	1992-2010	1.6*							1.6*
Female	1992-1994	-6.4	1994-2010	0.9*					0.9*
Female breast									
Incidence									
	1992-1999	1.3*	1999-2004	-2.2*	2004-2010	0.2			0.2
Death									
	1992-1995	-1.3*	1995-1998	-3.4*	1998-2010	-1.9*			-1.9*
Prostate									
Incidence									
	1992-1995	-11.2*	1995-2000	2.2	2000-2010	-2.0*			-2.0*
Death									
	1992-1994	-1.0	1994-2004	-3.8*	2004-2010	-3.1*			-3.1*

APC indicates annual percent change based on incidence (delay-adjusted) and mortality rates age adjusted to the 2000 US standard population; AAPC, average annual percent change.

*The APC or AAPC is significantly different from zero ($P < .05$).

Note: Trends analyzed by the Joinpoint Regression Program, version 4.0.3, allowing up to 3 joinpoints. Incidence trends based on Surveillance, Epidemiology, and End Results (SEER) 13 areas.

Figure 1 indicates the most common cancers expected to occur in men and women in 2014. Among men, cancers of the prostate, lung and bronchus, and colorectum will account for about 50% of all newly diagnosed cancers.

Prostate cancer alone will account for 27% (233,000) of incident cases in men. The 3 most commonly diagnosed types of cancer among women in 2014 will be breast, lung and bronchus, and colorectum, accounting for one-half of

all cases in women. Breast cancer alone is expected to account for 29% (232,670) of all new cancers among women.

Expected Numbers of Cancer Deaths

Table 1 also shows the expected numbers of deaths from cancer projected for 2014. It is estimated that about 585,720 Americans will die from cancer this year, corresponding to about 1,600 deaths per day. Cancers of the lung and bronchus, prostate, and colorectum in men and cancers of the lung and bronchus, breast, and colorectum in women continue to be the most common causes of cancer death. These 4 cancers account for almost half of the total cancer deaths among men and women (Fig. 1), with more than one-quarter of all cancer deaths due to lung cancer. Table 3 provides the estimated numbers of cancer deaths in 2014 by state for selected cancer sites.

Lifetime Probability of Developing Cancer

The lifetime probability of being diagnosed with an invasive cancer is higher for men (44%) than for women (38%) (Table 4). However, because of the earlier median age at diagnosis for breast cancer (61 years) compared with

prostate (66 years) and other major cancers, women have a slightly higher probability of developing cancer than men before age 65 years. These estimates are based on the average experience of the general population and may over- or underestimate individual risk because of differences in exposure (eg, smoking history), medical history, and/or genetic susceptibility.

Trends in Cancer Incidence

Figures 2 and 3 illustrate long-term trends in cancer incidence rates for all cancers combined and for selected cancer sites by sex. Trends since 1992 are presented in Table 5 in terms of the annual percent change in rates using joinpoint regression analysis. Joinpoint is a tool used to describe and quantify trends by fitting observed rates to lines connected at "joinpoints" where trends change in direction or magnitude.^{15,17}

The overall cancer incidence rate is 23% lower among women compared with men. However, during the past 5 years for which there are data (2006-2010), the incidence rate decreased by 0.6% per year among males but remained stable in females (Table 5). The decrease in men is driven by the rapid declines in colorectal (3.3% per year), prostate (2.0% per year), and lung (1.9% per year) cancers.

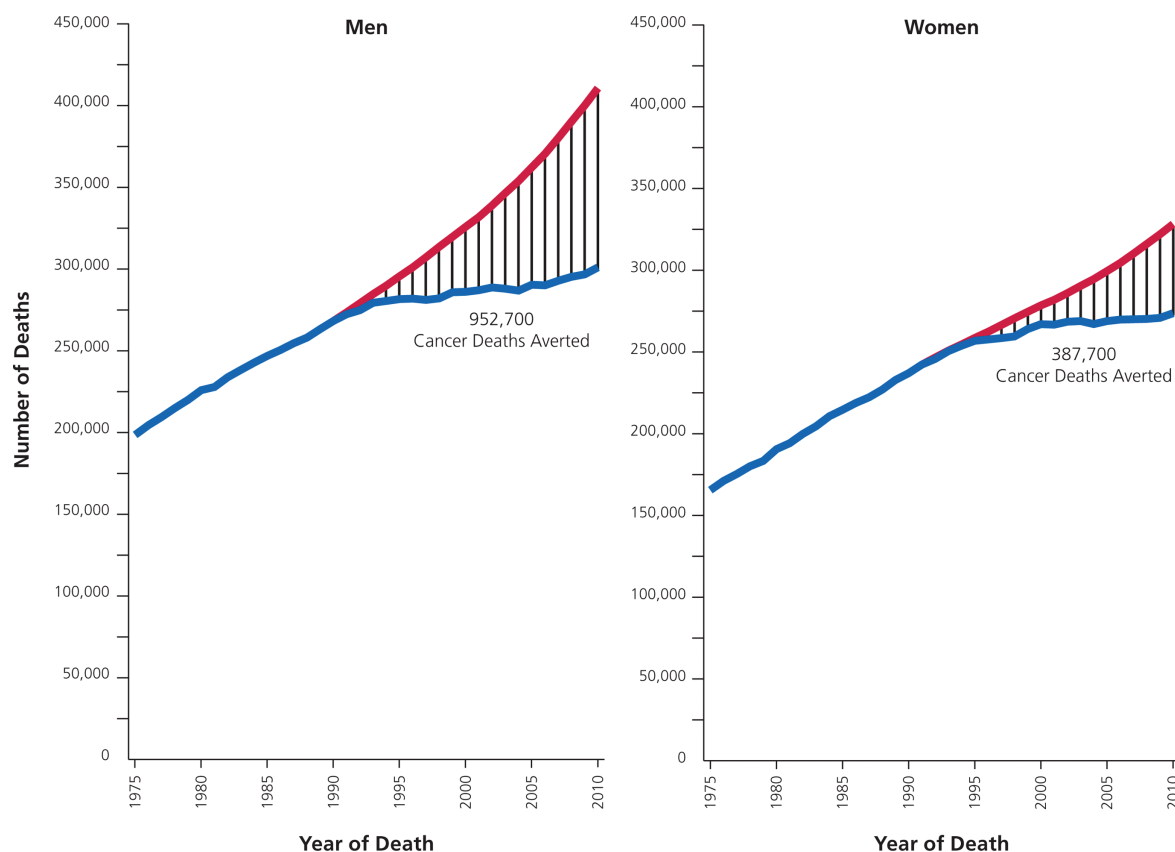


FIGURE 4. Total Number of Cancer Deaths Averted From 1991 to 2010 in Men and From 1992 to 2010 in Women.

The blue line represents the actual number of cancer deaths recorded in each year, and the red line represents the number of cancer deaths that would have been expected if cancer death rates had remained at their peak.

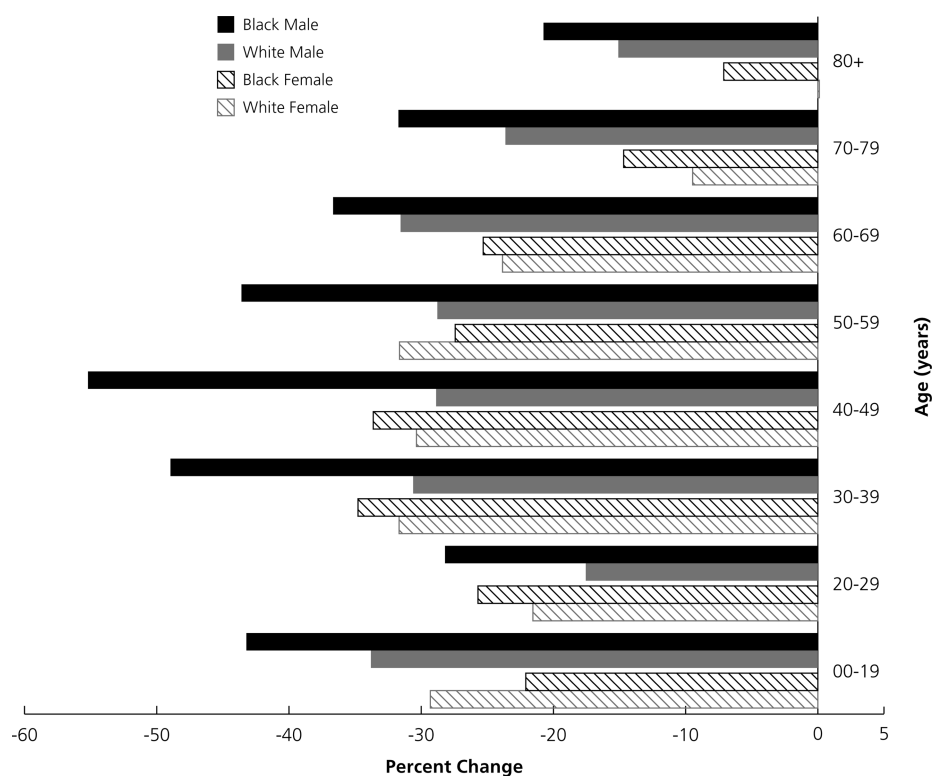


FIGURE 5. Relative Decline in Cancer Death Rates From 1991 to 2010 by Age, Race, and Sex.

The relative decline is the difference between the 2010 and 1991 rate expressed as a percentage of the 1991 rate.

Among women, although colorectal cancer declines are similar to those in men, the lung cancer rate has been slow to decline and breast cancer incidence rates have remained relatively flat since 2003 (Fig. 3).

Declines in incidence rates for the major cancers reflect improvements in cancer control and prevention. The long-term declines in colorectal cancer incidence rates since the mid-1980s have been attributed to both changes in risk factors and the introduction of screening.¹⁸ However, the rapid declines in recent years (eg, greater than 4.0% per year from 2008-2010) have been attributed to the increased uptake of colonoscopy, which prevents cancer by allowing for the removal of precancerous lesions.^{19,20} Prostate cancer incidence rates have been generally declining since around 2000, although rates have fluctuated widely from year to year (Fig. 3), likely reflecting variation in the prevalence of prostate-specific antigen testing for the detection of prostate cancer. Lung cancer incidence rates began declining in the mid-1980s in men and in the late 1990s in women as a result of reduced smoking prevalence.⁵ Differences in lung cancer incidence patterns between men and women (Fig. 3) reflect historical differences in tobacco use; cigarette smoking prevalence peaked about 20 years later in women than in men.²¹

In contrast to the stable or declining trends for most cancer types, incidence rates are increasing for melanoma of

the skin; esophageal adenocarcinoma; cancers of the thyroid, liver, kidney, anus, and pancreas; and human papillomavirus-positive oropharyngeal cancers.^{5,22} Among both men and women, the largest annual increases from 2006 to 2010 were for cancers of the thyroid (5.4% and 6.5% in men and women, respectively) and liver (3.7% and 2.9% in men and women, respectively) (Table 5).

Trends in Cancer Mortality

The overall cancer death rate rose for most of the 20th century, peaking at 215.1 deaths per 100,000 population in 1991. This increase was largely driven by rapid increases in lung cancer deaths among men as a consequence of the tobacco epidemic. Over the past 2 decades, however, there has been a steady decline in the cancer death rate (to 171.8 in 2010) as a result of advances in prevention, early detection, and treatment, including the implementation of comprehensive tobacco control. As a result of this 20% decline, an estimated 1,340,400 cancer deaths (952,700 in men and 387,700 in women) that would have occurred had peak rates persisted have been averted (Fig. 4).

Overall declines in the cancer death rate from 1991 to 2010 occurred among white women (16%), black women (20%), white men (24%), and black men (33%). Figure 5 illustrates the variation in the magnitude of the declines by

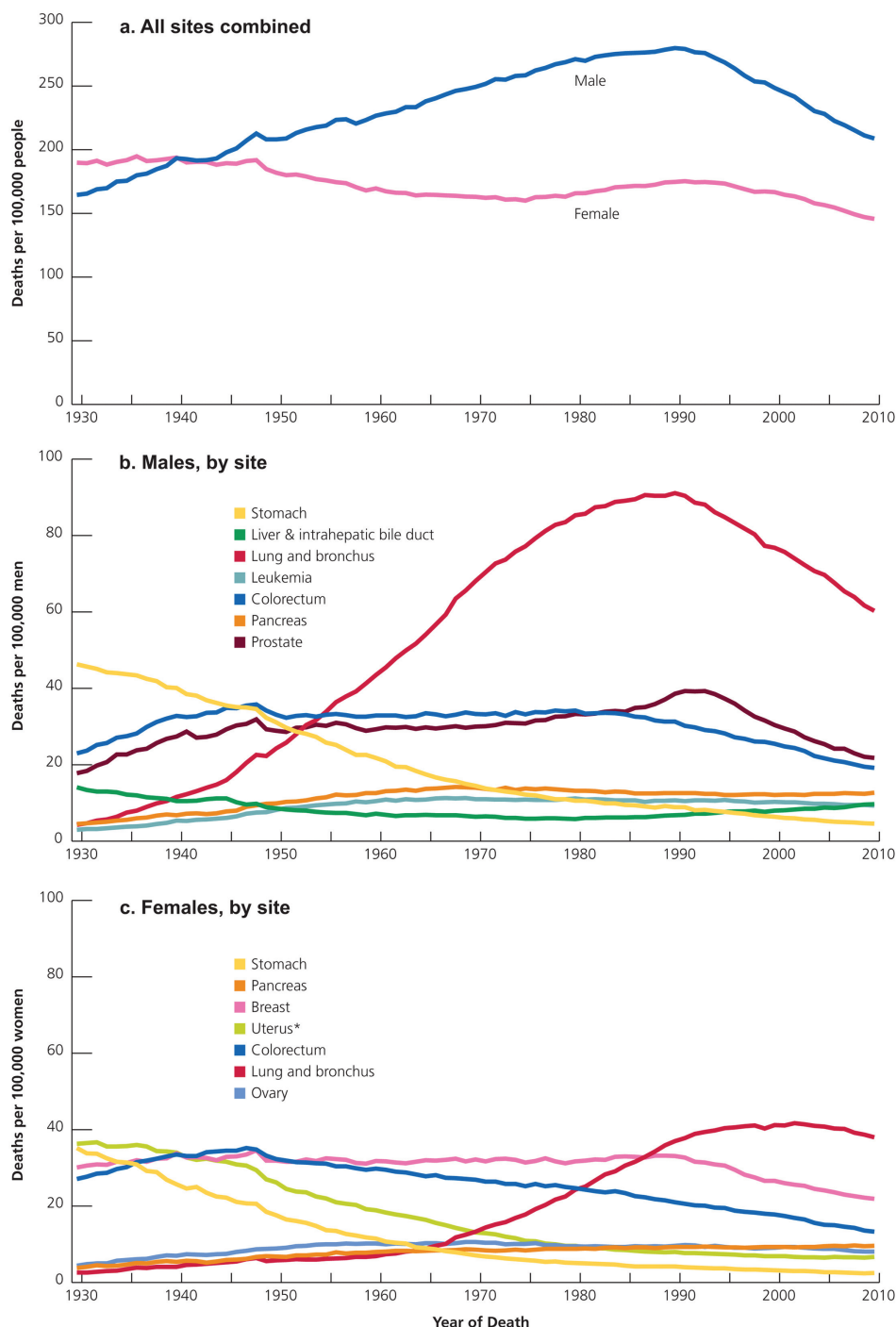


FIGURE 6. Trends in Death Rates Overall and for Selected Sites by Sex, United States, 1930 to 2010.

Rates are age adjusted to the 2000 US standard population. Due to changes in International Classification of Diseases (ICD) coding, numerator information has changed over time. Rates for cancers of the lung and bronchus, colorectum, liver, uterus, and ovary are affected by these changes. *Uterus includes uterine cervix and uterine corpus.

race, sex, and 10-year age group. Downturns occurred for black and white men and women of all ages with the exception of white women aged 80 years and older. Notably, black males experienced the largest drop in death rates in every age group. The sharpest decrease (44%–55%) occurred among black men aged 30 years to 59 years. In general, middle-aged men and women experienced the

largest declines, particularly compared with adults aged 70 years and older. The smaller declines among seniors reflect the lingering effects of the tobacco epidemic on older birth cohorts. Progress is more stunted among women than men in this generation because of the later onset and more protracted decline in smoking prevalence among women.

TABLE 6. Ten Leading Causes of Death by Age and Sex, United States, 2010

	ALL AGES		AGES 1 TO 19		AGES 20 TO 39		AGES 40 TO 59		AGES 60 TO 79		AGES ≥80	
	MALE All Causes 1,232,432	FEMALE All Causes 1,236,003	MALE All Causes 13,380	FEMALE All Causes 7,102	MALE All Causes 60,730	FEMALE All Causes 28,205	MALE All Causes 224,613	FEMALE All Causes 140,576	MALE All Causes 475,230	FEMALE All Causes 376,110	MALE All Causes 444,690	FEMALE All Causes 673,087
1	Heart diseases 307,384	Heart diseases 290,305	Accidents (unintentional injuries) 5,124	Accidents (unintentional injuries) 2,450	Accidents (unintentional injuries) 21,504	Accidents (unintentional injuries) 7,818	Cancer 54,440	Cancer 50,509	Cancer 156,723	Cancer 128,760	Heart diseases 131,682	Heart diseases 193,291
2	Cancer 301,037	Cancer 273,706	Assault (homicide) 1,978	Cancer 824	Intentional self-harm (suicide) 9,463	Cancer 4,429	Heart diseases 52,472	Heart diseases 21,458	Heart diseases 117,385	Heart diseases 72,817	Cancer 84,636	Cancer 89,153
3	Accidents (unintentional injuries) 75,921	Cerebro-vascular diseases 77,109	Intentional self-harm (suicide) 1,510	Congenital anomalies 501	Assault (homicide) 7,098	Heart diseases 2,339	Accidents (unintentional injuries) 23,991	Accidents (unintentional injuries) 11,469	Chronic lower respiratory diseases 31,964	Chronic lower respiratory diseases 31,397	Chronic lower respiratory diseases 27,899	Cerebro-vascular disease 51,984
4	Chronic lower respiratory diseases 65,423	Chronic lower respiratory diseases 72,657	Cancer 1,039	Assault (homicide) 500	Heart diseases 5,202	Intentional self-harm (suicide) 2,297	Intentional self-harm (suicide) 12,253	Chronic lower respiratory diseases 5,172	Cerebro-vascular disease 19,703	Cerebro-vascular disease 19,261	Cerebro-vascular disease 25,130	Alzheimer disease 50,503
5	Cerebro-vascular diseases 52,367	Alzheimer disease 58,130	Congenital anomalies 506	Intentional self-harm (suicide) 423	Cancer 4,163	Assault (homicide) 1,342	Chronic liver disease & cirrhosis 10,899	Cerebro-vascular disease 5,128	Diabetes mellitus 16,718	Diabetes mellitus 13,431	Alzheimer disease 19,934	Chronic lower respiratory diseases 35,719
6	Diabetes mellitus 35,490	Accidents (unintentional injuries) 44,938	Heart diseases 427	Heart diseases 259	HIV disease 1,016	Pregnancy, childbirth & puerperium 632	Diabetes mellitus 7,403	Chronic liver disease & cirrhosis 4,746	Accidents (unintentional injuries) 13,459	Nephritis, nephrotic syndrome & nephrosis 8,266	Influenza & pneumonia 13,266	Influenza & pneumonia 18,344
7	Intentional self-harm (suicide) 30,277	Diabetes mellitus 33,581	Chronic lower respiratory diseases 154	Influenza & pneumonia 96	Chronic liver disease & cirrhosis 828	Diabetes mellitus 587	Cerebro-vascular disease 6,675	Diabetes mellitus 4,445	Nephritis, nephrotic syndrome & nephrosis 9,581	Accidents (unintentional injuries) 7,937	Nephritis, nephrotic syndrome & nephrosis 12,217	Diabetes mellitus 15,082
8	Alzheimer disease 25,364	Influenza & pneumonia 26,482	Cerebro-vascular disease 133	Cerebro-vascular disease 93	Diabetes mellitus 779	Cerebro-vascular disease 576	Chronic lower respiratory diseases 5,027	Intentional self-harm (suicide) 3,793	Chronic liver disease & cirrhosis 7,734	Alzheimer disease 7,431	Accidents (unintentional injuries) 11,195	Nephritis, nephrotic syndrome & nephrosis 14,891
9	Nephritis, nephrotic syndrome & nephrosis 24,865	Nephritis, nephrotic syndrome & nephrosis 25,611	Influenza & pneumonia 114	Chronic lower respiratory diseases 90	Cerebro-vascular disease 661	HIV disease 553	HIV disease 4,054	Septicemia 2,269	Influenza & pneumonia 7,366	Septicemia 6,715	Diabetes mellitus 10,548	Accidents (unintentional injuries) 14,786
10	Influenza & pneumonia 23,615	Septicemia 18,743	In situ, benign, & unknown neoplasms 97	In situ, benign, & unknown neoplasms 90	Congenital anomalies 477	Chronic liver disease & cirrhosis 466	Viral hepatitis 3,306	Nephritis, nephrotic syndrome & nephrosis 2,087	Septicemia 6,768	Influenza & pneumonia 5,919	Parkinson disease 8,291	Hypertension & hypertensive renal disease* 10,756

HIV indicates human immunodeficiency virus.

*Includes primary and secondary hypertension.

Note: Deaths within each age group do not sum to all ages combined due to the inclusion of unknown ages. In accordance with the National Center for Health Statistics' cause-of-death ranking, "Symptoms, signs, and abnormal clinical or laboratory findings" and categories that begin with "Other" and "All other" were not ranked.

Source: US Final Mortality Data, 2010, National Center for Health Statistics, Centers for Disease Control and Prevention, 2013.

Figure 6 depicts long-term trends in cancer death rates among men and women overall and for selected cancer sites. In contrast to male cancer death rates, which rose continuously prior to 1990, female cancer death rates fell from the late 1940s to the mid-1970s (Fig. 6a). It is also interesting to note that prior to 1941, men had a lower risk

of dying from cancer than women due to the high death rate for uterine cancer (uterine corpus and uterine cervix combined), which was the leading cause of cancer death among women in the early 20th century. Uterine cancer death rates declined by more than 80% between 1930 and 2010, largely due to the widespread uptake of screening for

TABLE 7. Five Leading Types of Cancer Death by Age and Sex, United States, 2010

ALL AGES	<20	20 TO 39	40 TO 59	60 TO 79	≥80
MALE					
ALL SITES 300,464	ALL SITES 1,070	ALL SITES 4,163	ALL SITES 54,440	ALL SITES 156,723	ALL SITES 84,636
Lung & bronchus 87,699	Leukemia 293	Leukemia 550	Lung & bronchus 14,687	Lung & bronchus 52,233	Lung & bronchus 20,504
Prostate 28,561	Brain & ONS 272	Brain & ONS* 503	Colorectum 5,645	Colorectum 13,272	Prostate 15,188
Colorectum 27,074	Bones & joints 92	Colorectum 442	Liver & bile duct 4,458	Prostate 11,949	Colorectum 7,708
Pancreas 18,699	Soft tissue (including heart) 83	Non-Hodgkin lymphoma 276	Pancreas 3,748	Pancreas 10,371	Urinary bladder 4,766
Liver & intrahepatic bile duct 13,658	Non-Hodgkin lymphoma 55	Lung & bronchus 266	Esophagus 2,737	Liver & intrahepatic bile duct 6,803	Pancreas 4,466
FEMALE					
ALL SITES 273,706	ALL SITES 855	ALL SITES 4,429	ALL SITES 50,509	ALL SITES 128,760	ALL SITES 89,153
Lung & bronchus 70,550	Leukemia 235	Breast 963	Breast 11,507	Lung & bronchus 39,540	Lung & bronchus 19,681
Breast 40,996	Brain & ONS* 226	Uterine cervix 430	Lung & bronchus 11,082	Breast 17,375	Breast 11,151
Colorectum 24,972	Bones & joints 74	Leukemia 395	Colorectum 4,185	Colorectum 9,855	Colorectum 10,588
Pancreas 18,189	Soft tissue 63	Colorectum 338	Ovary 3,113	Pancreas 8,911	Pancreas 6,692
Ovary 14,572	Liver & intrahepatic bile duct 25	Brain & ONS* 309	Pancreas 2,506	Ovary 7,338	Non-Hodgkin lymphoma 4,144

ONS indicates other nervous system.

Note: Ranking order excludes category titles that begin with "Other."

the prevention and early detection of cervical cancer. A similar dramatic decline occurred for stomach cancer, which accounted for 30% and 20% of male and female cancer deaths, respectively, in the 1930s. By 2010, stomach cancer accounted for just 2% of cancer deaths. Rates of stomach cancer have declined worldwide due to improved hygiene, resulting in a lower prevalence of *Helicobacter pylori*, and advances in food preservation techniques (eg, refrigeration), which have led to lower salt intake and higher consumption of fresh fruits and vegetables.

During the most recent 5 years for which data are available, the average annual decline in cancer death rates was slightly larger among men (1.8%) than women (1.4%) (Fig. 6) (Table 5). These declines are driven by continued decreases in death rates for the 4 major cancer sites (Fig. 6). Due to the reduction in tobacco use over the past 50 years, the lung cancer death rate declined 34% between 1991 and 2010 among males and 9% between 2002 and 2010 among females.^{5,23} Death rates for breast, prostate, and colorectal cancers are down from peak rates by 34%,

45%, and 46%, respectively, as a result of improvements in early detection and treatment.^{5,18,24,25}

In contrast, joinpoint trend analysis for 2001 to 2010 indicates that death rates are rising for cancers of the oropharynx, anus, liver, pancreas, and soft tissue (including the heart), and for melanoma in men.⁵ The rate of death from thyroid cancer is also increasing according to joinpoint analysis (Table 5), although the observed rates increased only slightly from 0.47 (per 100,000 population) to 0.50 among men and from 0.48 to 0.51 among women.

Recorded Number of Deaths in 2010

A total of 2,468,435 deaths were recorded in the United States in 2010, of which 574,743 (23%) were from cancer. Overall, cancer is the second leading cause of death following heart disease. However, within 20-year age groups, cancer is the leading cause of death among adults aged 40 years to 79 years (Table 6). Among females, cancer is the first or second leading cause of death in every age group.

TABLE 8. Incidence Rates for Selected Cancers by State, United States, 2006 to 2010

STATE	ALL CANCERS		BREAST	COLORECTUM		LUNG & BRONCHUS		NON-HODGKIN LYMPHOMA		PROSTATE	URINARY	BLADDER
	MALE	FEMALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	MALE	FEMALE
Alabama	573.2	395.2	118.7	57.2	40.3	103.2	54.0	19.5	13.6	157.7	33.3	7.5
Alaska	521.0	430.3	127.7	53.1	44.5	83.8	60.4	21.8	16.1	137.3	38.0	9.8
Arizona	441.4	371.2	110.2	41.4	32.0	61.8	47.9	18.0	13.4	112.7	32.2	8.4
Arkansas*†	557.7	388.1	110.3	55.6	40.5	108.4	60.4	21.2	15.3	156.4	33.9	8.1
California	505.7	397.1	122.0	49.4	37.3	60.4	44.4	22.9	15.6	140.3	33.5	8.0
Colorado	483.1	396.4	125.3	43.6	33.6	56.1	44.2	22.5	15.9	142.7	31.2	8.2
Connecticut	576.2	456.9	136.3	51.3	39.2	75.5	59.5	25.2	17.3	160.0	46.8	12.6
Delaware	601.7	443.3	126.5	53.1	39.6	87.1	65.8	23.7	16.7	177.3	43.2	11.3
Dist. of Columbia	574.8	427.7	139.7	50.9	44.8	77.5	48.1	21.8	13.4	194.4	25.1	9.0
Florida	518.8	399.9	114.3	47.8	36.6	79.4	56.7	21.9	15.2	131.2	35.3	8.6
Georgia	568.7	403.1	121.5	52.1	38.4	93.3	55.0	21.5	14.8	165.7	33.9	7.9
Hawaii	484.4	393.4	123.1	56.9	38.0	64.3	38.7	20.7	13.7	119.9	25.8	6.3
Idaho	513.3	410.1	119.5	43.2	34.8	61.5	47.2	21.3	17.5	152.9	36.1	8.9
Illinois	566.6	440.3	126.4	59.1	43.4	86.7	60.9	24.0	16.4	153.9	39.4	9.9
Indiana	527.4	422.0	117.4	54.7	41.8	96.4	63.5	23.3	17.0	122.4	35.2	8.8
Iowa	555.7	437.1	123.4	56.9	44.3	84.7	54.9	26.8	18.9	137.2	41.7	8.8
Kansas	552.2	422.0	123.2	55.0	39.9	80.2	54.5	23.4	16.8	152.4	37.9	9.3
Kentucky	611.2	462.4	121.3	63.9	46.0	125.9	80.3	25.3	17.5	134.6	40.5	9.7
Louisiana	603.4	413.6	119.7	62.4	44.0	99.6	57.7	24.5	16.5	169.3	34.1	8.1
Maine	581.5	460.6	126.5	51.4	41.2	91.5	67.3	25.4	17.9	144.8	48.0	13.2
Maryland	529.1	415.0	128.0	47.6	36.7	74.4	55.8	21.3	15.0	157.2	33.4	9.2
Massachusetts	568.1	460.4	134.2	49.9	38.8	78.3	64.1	25.0	16.5	153.6	43.9	11.9
Michigan	574.1	433.5	120.0	51.0	39.3	86.1	61.6	24.9	17.7	163.7	41.3	10.7
Minnesota‡	-	-	-	-	-	-	-	-	-	-	-	-
Mississippi	598.6	396.9	113.8	61.3	44.7	112.7	56.2	21.6	14.5	166.3	30.8	7.2
Missouri	530.8	423.3	121.5	55.3	40.8	95.6	64.3	22.3	16.0	126.2	34.8	8.4
Montana	520.8	421.9	124.1	51.1	39.0	68.9	56.3	22.8	15.0	155.2	36.4	9.9
Nebraska	526.2	420.9	122.3	57.7	44.4	74.6	51.2	23.5	17.7	143.0	35.0	8.6
Nevada*§	509.8	399.2	112.7	52.0	38.1	75.7	64.7	20.4	15.3	138.9	37.3	10.7
New Hampshire	580.3	452.1	132.0	46.7	38.0	80.1	62.2	25.1	17.1	155.4	49.1	13.2
New Jersey	582.6	450.6	129.3	54.7	41.2	72.8	55.3	25.2	17.8	169.2	43.6	11.6
New Mexico	461.9	362.5	108.8	44.1	33.6	52.9	38.1	18.2	13.8	134.1	26.2	6.2
New York	585.4	449.2	127.7	53.3	40.9	76.3	56.0	26.3	18.1	167.3	42.1	10.6
North Carolina	564.9	416.0	124.9	50.8	37.1	96.7	57.2	22.6	15.7	151.9	37.5	8.9
North Dakota	528.6	410.2	123.0	59.2	41.8	68.1	43.3	22.0	18.4	156.2	38.6	9.2
Ohio*§	548.1	425.4	120.5	55.5	42.2	92.6	60.7	23.0	16.1	145.2	38.7	9.6
Oklahoma	552.2	422.0	121.7	53.6	40.8	96.1	62.7	22.4	17.1	148.4	34.7	8.6
Oregon	508.8	429.2	129.5	45.5	36.8	70.6	57.6	22.8	15.3	139.3	37.1	9.5
Pennsylvania	573.6	454.8	126.0	56.0	42.8	84.4	57.9	25.2	17.8	149.5	44.1	11.0
Rhode Island	575.7	462.4	131.0	51.3	41.3	84.1	64.5	22.5	17.9	148.9	48.7	13.8
South Carolina	551.7	401.1	122.3	50.3	37.9	94.1	53.9	20.5	13.4	152.8	30.8	8.4
South Dakota	499.4	395.9	117.9	55.7	41.8	73.8	48.0	21.6	15.9	145.5	33.6	8.3
Tennessee	562.0	416.0	118.8	53.4	40.4	103.4	61.3	22.6	16.3	144.3	35.4	8.2
Texas	513.9	389.9	114.4	50.9	35.7	78.2	49.0	22.2	15.9	133.2	29.5	6.9
Utah	494.8	357.7	110.8	39.4	31.2	34.1	23.3	25.2	16.3	175.9	31.4	5.3
Vermont	539.8	453.8	131.4	44.8	38.3	81.2	65.6	24.1	17.7	139.9	40.5	12.2
Virginia*	521.7	397.4	124.5	46.9	36.7	82.2	53.9	21.7	14.2	150.3	33.3	8.3
Washington	544.6	437.7	131.0	46.1	36.3	72.1	57.3	26.2	17.4	151.6	38.9	9.3
West Virginia	557.7	434.1	110.2	58.0	43.8	106.4	70.0	23.2	17.5	131.8	38.8	10.7
Wisconsin	530.8	419.2	122.5	48.4	37.5	73.6	53.4	24.3	17.3	142.9	39.2	9.7
Wyoming	497.1	384.2	110.8	47.5	37.7	57.3	46.0	19.8	15.0	152.2	38.7	9.9
United States	542.3	418.8	122.2	51.7	39.1	80.0	55.1	23.3	16.3	146.6	36.9	9.1

Rates are per 100,000 and age adjusted to the 2000 US standard population.

*This state's data are not included in the US combined rates because it did not meet high-quality standards for one or more years during 2006 to 2010 according to the North American Association of Central Cancer Registries (NAACCR).

†Rates are based on incidence data for 2006 to 2008.

‡This state's registry did not submit cancer incidence data to the NAACCR.

§Rates are based on incidence data for 2006 to 2009.

Table 7 presents the number of deaths from all cancers combined and from the 5 most common sites for each 20-year age group by sex. For all ages combined, men have a

higher number of deaths for shared sites with the exception of pancreatic cancer, for which the burden is similar. Among males, leukemia is the leading cause of cancer death in those

TABLE 9. Death Rates for Selected Cancers by State, United States, 2006 to 2010

STATE	ALL SITES		BREAST	COLORECTUM		LUNG & BRONCHUS		NON-HODGKIN LYMPHOMA		PANCREAS		PROSTATE
	MALE	FEMALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE
Alabama	255.4	156.4	23.3	22.5	15.0	87.4	40.8	8.0	5.3	13.2	9.7	28.9
Alaska	210.5	161.0	24.2	19.6	15.1	61.8	45.8	7.6	5.5	12.9	9.8	22.5
Arizona	186.0	131.7	20.6	17.1	11.9	49.8	33.2	7.6	4.8	11.4	8.3	20.5
Arkansas	250.6	160.9	23.4	22.7	15.6	90.1	45.4	8.6	5.5	13.2	9.6	25.3
California	189.8	139.8	21.9	17.5	12.7	47.2	32.3	7.8	4.9	11.7	9.4	22.3
Colorado	181.4	133.2	19.6	16.7	12.4	44.2	31.3	7.8	4.4	10.8	9.0	22.9
Connecticut	199.5	144.7	21.6	16.2	12.1	52.7	38.2	7.5	4.9	13.9	9.9	22.3
Delaware	228.0	159.3	22.8	20.4	13.3	68.2	47.7	8.3	4.8	13.0	9.5	23.7
Dist. of Columbia	248.5	167.6	29.8	22.2	18.9	61.9	35.0	8.0	4.0	16.9	12.0	38.8
Florida	204.5	140.7	21.3	18.1	12.8	61.4	38.8	7.8	4.9	12.1	8.8	20.1
Georgia	228.3	147.8	23.4	20.2	13.8	73.1	38.7	7.6	4.5	12.5	9.1	27.1
Hawaii	178.4	117.1	16.4	17.3	10.9	48.8	25.9	7.2	4.3	12.3	9.7	15.7
Idaho	192.9	140.2	21.6	16.1	12.4	49.2	34.5	7.2	5.6	12.0	9.2	25.9
Illinois	223.3	158.5	23.6	21.7	15.4	65.8	41.9	8.5	5.3	13.0	10.1	24.3
Indiana	238.8	161.5	23.9	21.3	14.5	79.4	46.7	9.3	5.4	13.0	9.4	23.2
Iowa	212.6	148.4	21.3	20.0	15.2	64.1	38.3	8.9	5.5	12.2	8.8	22.7
Kansas	214.4	146.5	22.2	20.6	13.1	66.8	39.4	9.1	5.3	12.6	9.1	20.7
Kentucky	262.7	173.3	23.1	23.4	16.2	97.1	55.8	9.0	5.7	12.7	9.6	23.9
Louisiana	254.7	164.4	25.4	24.0	15.7	80.9	43.6	8.6	5.2	14.2	11.1	26.6
Maine	231.1	158.3	20.9	20.2	13.9	69.9	46.0	8.8	5.1	11.6	9.8	22.9
Maryland	216.9	154.0	24.5	20.5	14.0	61.7	40.8	7.4	4.5	12.9	10.3	25.0
Massachusetts	216.6	152.5	21.3	18.7	13.4	60.6	42.1	7.8	4.8	12.7	10.4	22.4
Michigan	223.0	159.5	24.0	19.4	14.2	68.2	43.9	8.9	5.8	13.9	10.2	21.8
Minnesota	203.9	145.0	20.9	17.6	12.6	53.3	37.0	9.4	5.2	11.9	9.3	23.9
Mississippi	270.6	159.1	24.7	24.8	16.5	95.4	42.0	8.3	4.7	14.1	10.1	31.2
Missouri	231.1	160.2	24.2	21.2	14.6	76.5	46.2	8.3	5.4	13.0	9.9	21.8
Montana	195.3	145.1	19.8	16.7	14.0	52.1	39.6	7.9	4.9	11.8	8.4	26.4
Nebraska	208.4	143.8	20.1	21.4	15.0	60.3	35.5	8.4	5.7	12.0	9.7	22.9
Nevada	211.4	153.6	23.0	21.0	14.9	59.7	46.5	7.1	4.6	12.6	9.7	23.9
New Hampshire	211.7	154.2	21.3	17.9	13.1	59.0	43.3	7.3	4.7	13.2	10.4	22.4
New Jersey	207.8	154.3	25.2	21.0	14.9	55.8	37.2	7.8	5.2	13.4	10.1	21.6
New Mexico	187.4	131.6	20.9	18.8	12.7	43.4	28.6	6.6	4.5	10.9	8.4	24.4
New York	199.7	145.4	22.3	19.2	13.8	54.3	36.1	7.9	4.9	12.8	9.9	21.8
North Carolina	232.2	149.9	23.1	19.4	13.1	76.6	40.7	7.5	5.0	12.0	9.6	25.8
North Dakota	203.6	137.2	21.6	21.2	13.9	54.1	32.2	6.7	5.1	12.6	8.3	23.6
Ohio	236.5	162.6	24.8	21.8	15.2	74.8	44.2	9.2	5.6	13.1	10.1	23.8
Oklahoma	238.8	161.0	23.9	22.1	14.5	79.6	46.6	9.0	6.0	12.2	8.9	23.5
Oregon	208.5	153.4	21.6	18.1	13.3	58.4	42.7	8.5	5.3	12.3	9.9	24.4
Pennsylvania	225.6	157.0	23.8	21.5	15.0	65.8	39.6	8.9	5.5	13.4	10.0	22.7
Rhode Island	221.8	149.6	20.8	18.6	13.2	64.9	42.6	8.3	4.4	12.2	8.9	20.6
South Carolina	240.1	151.7	23.5	20.2	14.2	77.1	39.9	7.8	4.7	12.7	9.8	27.5
South Dakota	206.8	141.3	20.3	19.8	13.9	61.9	34.1	7.8	5.3	11.1	9.3	23.4
Tennessee	256.4	160.4	23.3	21.8	15.3	89.5	46.6	9.2	5.4	13.1	9.4	25.2
Texas	209.3	141.8	21.8	20.0	13.1	60.7	35.4	7.9	4.9	11.8	8.8	21.3
Utah	157.1	112.3	21.8	14.3	10.7	27.5	16.8	7.7	4.7	10.0	8.4	25.9
Vermont	212.0	154.3	20.4	17.4	14.8	62.3	45.1	7.8	5.0	12.4	9.3	22.1
Virginia	221.1	151.4	24.0	19.2	14.0	67.1	40.0	8.2	4.8	12.7	9.7	24.7
Washington	207.2	152.2	21.5	17.1	12.5	57.0	42.1	8.6	5.4	12.7	10.0	24.2
West Virginia	246.9	168.6	22.1	23.4	16.1	84.3	50.9	8.7	6.6	11.3	7.8	20.9
Wisconsin	213.7	149.7	21.3	18.0	12.7	58.4	38.4	9.0	5.5	12.8	10.0	24.5
Wyoming	196.0	146.8	21.3	18.7	14.7	49.9	36.4	7.6	5.3	12.9	9.0	22.8
United States	215.3	149.7	22.6	19.6	13.9	63.5	39.2	8.2	5.1	12.5	9.6	23.0

Rates are per 100,000 and age adjusted to the 2000 US standard population.

aged younger than 40 years, while lung cancer ranks first in men aged 40 years and older. Among females, leukemia is the leading cause of cancer death in children and adolescents (aged younger than 20 years), breast cancer ranks first in women aged 20 years to 59 years, and lung cancer causes the most cancer deaths in those aged 60 years and older.

Regional Variations in Cancer Rates

Tables 8 and 9 depict cancer incidence and death rates for selected cancers by state. Lung cancer shows the largest geographic variation in cancer occurrence by far, reflecting the large historical and continuing differences in smoking prevalence among states.²³ For example, lung cancer

TABLE 10. Incidence and Death Rates by Site, Race, and Ethnicity, United States, 2006 to 2010

	NON-HISPANIC WHITE	BLACK	ASIAN AMERICAN/ PACIFIC ISLANDER	AMERICAN INDIAN/ ALASKA NATIVE*	HISPANIC
Incidence					
All sites					
Male	548.6	601.0	326.1	441.1	426.8
Female	436.2	395.9	282.6	372.0	330.8
Breast (female)	127.3	118.4	84.7	90.3	91.1
Colorectum					
Male	50.9	62.5	40.8	51.7	47.3
Female	38.6	46.7	31.0	42.7	32.6
Kidney & renal pelvis					
Male	21.6	23.0	10.6	30.6	20.5
Female	11.2	12.2	5.1	17.5	11.5
Liver & intrahepatic bile duct					
Male	8.7	14.9	21.3	17.8	18.8
Female	2.9	4.4	8.0	8.0	6.9
Lung & bronchus					
Male	82.9	94.7	48.8	70.2	45.9
Female	59.9	50.4	28.0	52.1	26.6
Prostate	138.6	220.0	75.0	104.1	124.2
Stomach					
Male	7.8	15.7	15.6	13.1	13.9
Female	3.5	8.1	9.0	6.9	8.2
Uterine cervix	7.2	10.3	6.7	9.7	10.9
Mortality					
All sites					
Male	217.3	276.6	132.4	191.0	152.2
Female	153.6	171.2	92.1	139.0	101.3
Breast (female)	22.7	30.8	11.5	15.5	14.8
Colorectum					
Male	19.2	28.7	13.1	18.7	16.1
Female	13.6	19.0	9.7	15.4	10.2
Kidney & renal pelvis					
Male	5.9	5.7	3.0	9.5	5.1
Female	2.6	2.6	1.2	4.4	2.3
Liver & intrahepatic bile duct					
Male	7.1	11.8	14.4	13.2	12.3
Female	2.9	4.1	6.0	6.1	5.4
Lung & bronchus					
Male	65.7	78.5	35.5	49.6	31.3
Female	42.7	37.2	18.4	33.1	14.1
Prostate	21.3	50.9	10.1	20.7	19.2
Stomach					
Male	3.9	9.8	8.7	8.1	7.6
Female	1.9	4.7	5.1	3.8	4.4
Uterine cervix	2.1	4.2	1.9	3.5	2.9

Rates are per 100,000 population and age adjusted to the 2000 US standard population. Nonwhite race categories are not mutually exclusive of Hispanic origin.

*Data based on Indian Health Service Contract Health Service Delivery Areas.

incidence rates in Kentucky, which has historically had the highest smoking prevalence, are almost 4-fold higher than those in Utah, which has the lowest smoking prevalence (126 vs 34 cases per 100,000 men). There is also a large range for prostate cancer incidence rates, from 112.7 in Arizona to 194.4 in the District of Columbia, which likely reflects state differences in PSA testing prevalence and racial distribution.²⁶ In contrast, state variations for other cancer sites are smaller in both absolute and relative terms. For example, the breast cancer incidence rate ranges from

140 (per 100,000 population) in the District of Columbia to 109 in New Mexico, a difference of 28%. State variation in incidence rates reflects differences in the use of screening tests or detection practices in addition to differences in disease occurrence.

Cancer Occurrence by Race/Ethnicity

Cancer incidence and death rates vary considerably between and within racial and ethnic groups. Of the 5 broadly

TABLE 11. Average Annual Percent Change in Cancer Incidence and Mortality Rates From 2001 to 2010 by Race/Ethnicity, United States

	INCIDENCE		MORTALITY	
	MALE	FEMALE	MALE	FEMALE
All race/ethnicities	-1.6*	-0.3*	-1.8*	-1.4*
Non-Hispanic white†	-1.6*	-0.5*	-1.6*	-1.3*
Black	-1.8*	-0.1	-2.5*	-1.7*
Asian American/Pacific Islander	-1.7*	0.0	-1.6*	-1.0*
American Indian/Alaska Native‡	-0.8	-0.2	-0.1	0.1
Hispanic†	-2.0*	-0.8*	-1.6*	-1.4*

*Average annual percent change is statistically significant ($P < .05$).

†Excludes deaths from Connecticut, District of Columbia, Maine, Maryland, Minnesota, New Hampshire, New York, North Dakota, Oklahoma, South Carolina, and Vermont due to unreliable Hispanic origin data for some years.

‡Data based on Indian Health Service Contract Health Service Delivery Areas.

Notes: Trends analyzed by the Joinpoint Regression Program, version 3.5.0, allowing up to 2 joinpoints. Incidence trends based on the North American Association of Central Cancer Registries (NAACCR) data. Nonwhite race categories are not mutually exclusive of Hispanic origin.

defined groups in Table 10, black men have the highest cancer incidence and death rates—about double those of Asian Americans, who have the lowest rates. Cancer incidence and death rates are higher among black than white men for every site included in Table 10 except kidney cancer mortality. Factors known to contribute to racial disparities vary by cancer site and include differences in exposure to underlying risk factors (eg, historical smoking prevalence for lung cancer), access to high-quality screening (breast, cervical, and colorectal cancers), and timely diagnosis and treatment.²⁷ It is notable that although white women have the highest breast cancer incidence rate, black women have the highest breast cancer mortality rate. The higher incidence rate among white women is thought to reflect a combination of factors that affect both diagnosis (more prevalent mammography) and underlying disease occurrence (such as later age at first birth and greater use of menopausal hormone therapy).²⁸ The high breast cancer mortality rate among black women has been attributed to a higher prevalence of comorbidities, a longer time to follow-up after an abnormal mammogram, less access to high-quality treatment, and a higher prevalence of aggressive tumor characteristics.^{29–31}

Cancer incidence and death rates are lower among Asian Americans/Pacific Islanders, American Indians/Alaska Natives, and Hispanics than whites for all cancer sites combined and for the 4 most common cancer sites. However, cancers associated with infectious agents (eg, those of the uterine cervix, stomach, and liver) are generally more common in nonwhite populations. For example, stomach and liver cancer incidence and death rates are twice as high in the Asian American/Pacific Islander population as in whites, reflecting a higher prevalence of chronic infection with *Helicobacter pylori* and hepatitis B virus, respectively.³² Kidney cancer incidence and death

rates are the highest among American Indians/Alaska Natives, which may be due in part to high rates of obesity and smoking in this population.³³

Table 11 presents trends in cancer incidence and death rates during the most recent 10 years for which there are data (2001–2010) by race and ethnicity. These trends are based on rates that are not adjusted for reporting delays because the long-term incidence data required for delay adjustment are not available for populations other than whites or blacks. Among men, incidence rates declined between 1.6% and 2.0% per year for all groups except American Indians/Alaska Natives. Declines in death rates among men were of a similar magnitude, and were highest among blacks (2.5% per year). Among women, incidence rate declines were smaller and were confined to non-Hispanic whites (0.5% per year) and Hispanics (0.8% per year), though mortality declines of greater than 1.0% per year were experienced by all racial/ethnic groups except American Indians/Alaska Natives. As in men, black women had the largest annual decline in cancer death rates (1.7%).

Cancer Survival

The stage-specific 5-year relative survival rate is lower for blacks than for whites for nearly every cancer type (Fig. 7). Studies suggest that racial disparities in survival are primarily due to differences in treatment, stage at diagnosis, and comorbidities, as opposed to differences in cancer biology.³⁴ As shown in Figure 8, blacks are less likely than whites to be diagnosed with cancer at a localized stage, when treatment is more successful. This disparity is particularly striking for cancers of the breast, cervix, uterine corpus, and oral cavity and pharynx. Lower socioeconomic status among blacks likely explains much of the stage disparity and is also associated with receipt of less

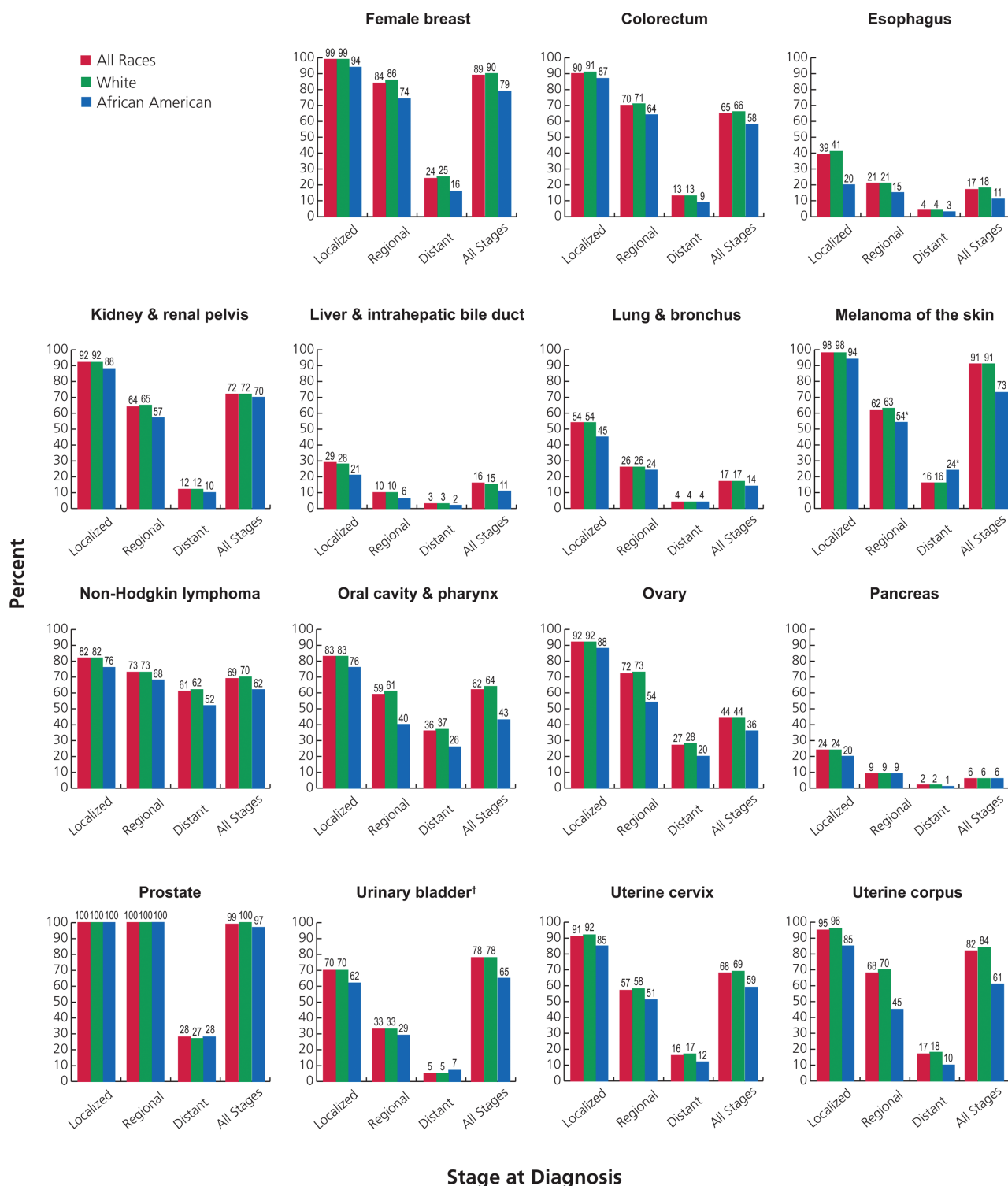


FIGURE 7. Five-Year Relative Survival Rates for Selected Cancers by Race and Stage at Diagnosis, United States, 2003 to 2009.

*The standard error of the survival rate is between 5 and 10 percentage points.

†The survival rate for carcinoma in situ of the urinary bladder is 96% for All Races, 97% for Whites, and 91% for blacks.

aggressive treatment.^{35,36} Even among Medicare-insured patients, blacks are less likely than whites to receive standard cancer therapies for lung, breast, colorectal, and prostate cancers.³⁷

There have been notable improvements in survival over the past 3 decades for most cancers among both whites and blacks (Table 12). Between 1975 and 1977 and 2003 and 2009, overall 5-year relative survival rates increased 19

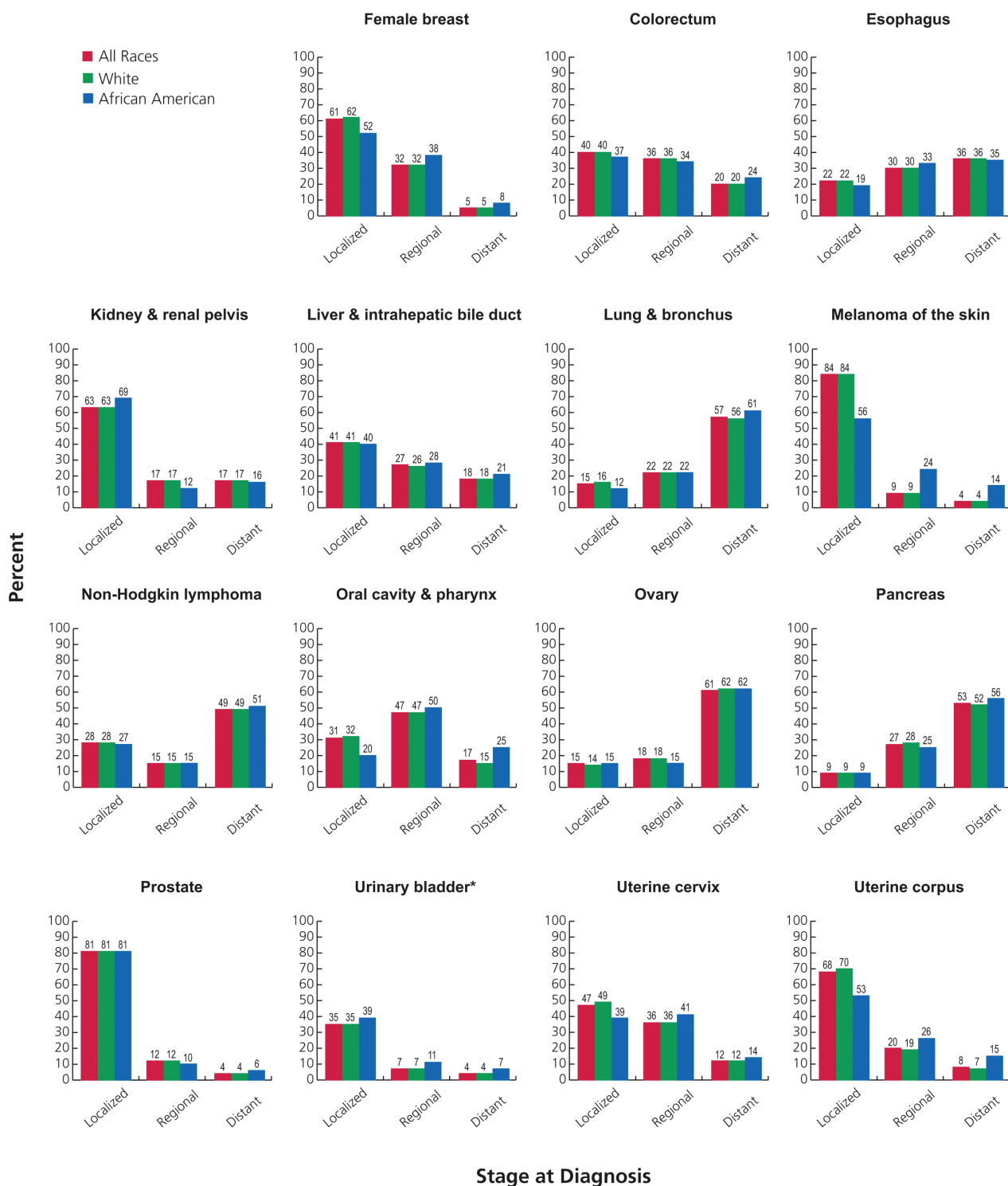


FIGURE 8. Stage Distribution of Selected Cancers by Race, United States, 2003 to 2009.

*The proportions of carcinoma in situ of the urinary bladder are 51% for All Races, 52% for Whites, and 38% for blacks. Stage categories do not sum to 100% because sufficient information is not available to assign a stage to all cancer cases.

percentage points among whites and 22 percentage points among blacks. The largest improvements in survival have been for leukemia and non-Hodgkin lymphoma, while lung and pancreatic cancers have shown the least improvement.

Relative survival rates cannot be calculated for some minority populations because accurate life tables are not available. Comparison of cause-specific survival rates for patients diagnosed from 2003 to 2009 indicates that Asian American/Pacific Islander women have the highest

TABLE 12. Trends in 5-Year Relative Survival Rates* (%) by Race and Year of Diagnosis, United States, 1975 to 2009

	ALL RACES			WHITE			BLACK		
	1975 TO 1977	1987 TO 1989	2003 TO 2009	1975 TO 1977	1987 TO 1989	2003 TO 2009	1975 TO 1977	1987 TO 1989	2003 TO 2009
All sites	49	55	68†	50	57	69†	39	43	61†
Brain & other nervous system	22	29	35†	22	28	33†	25	32	41†
Breast (female)	75	84	90†	76	85	92†	62	71	79†
Colon	51	60	65†	51	61	67†	45	52	56†
Esophagus	5	9	19†	6	11	20†	4	7	14†
Hodgkin lymphoma	72	79	88†	72	80	89†	70	72	83†
Kidney & renal pelvis	50	57	73†	50	57	73†	49	55	72†
Larynx	66	66	63†	67	67	64	58	56	52
Leukemia	34	43	59†	35	44	60†	33	35	53†
Liver & intrahepatic bile duct	3	5	18†	3	6	17†	2	3	12†
Lung & bronchus	12	13	18†	12	13	18†	11	11	14†
Melanoma of the skin	82	88	93†	82	88	93†	57‡	79‡	77‡
Myeloma	25	27	45†	24	27	45†	30	30	44†
Non-Hodgkin lymphoma	47	51	71†	47	51	72†	48	46	64†
Oral cavity	53	54	65†	54	56	67†	36	34	46†
Ovary	36	38	44†	35	38	44†	42	34	36
Pancreas	2	4	6†	3	3	7†	2	6	6†
Prostate	68	83	100†	69	84	100†	61	71	98†
Rectum	48	58	68†	48	59	68†	44	52	62†
Stomach	15	20	29†	14	18	28†	16	19	29†
Testis	83	95	97†	83	96	97†	73‡§	88‡	90
Thyroid	92	94	98†	92	94	98†	90	92	97†
Urinary bladder	72	79	80†	73	80	81†	50	63	64†
Uterine cervix	69	70	69	70	73	71	64	57	63
Uterine corpus	87	82	84†	88	84	86†	60	57	64

*Survival rates are adjusted for normal life expectancy and are based on cases diagnosed in the Surveillance, Epidemiology, and End Results (SEER) 9 areas from 1975 to 1977, 1987 to 1989, and 2003 to 2009, all followed through 2010.

†The difference in rates between 1975 to 1977 and 2003 to 2009 is statistically significant ($P < .05$).

‡The standard error of the survival rate is between 5 and 10 percentage points.

§Survival rate is for 1978 to 1980.

probability of surviving 5 years after a cancer diagnosis (69.5%), while American Indian/Alaska Native men have the lowest probability of surviving (56.8%).⁵ For both sexes combined, whites and Hispanics have the highest 5-year cause-specific survival (66.7%), while American Indians/Alaska Natives have the lowest (59.0%).

Limitations

The projected numbers of new cancer cases and cancer deaths should be interpreted with caution because they are model-based estimates that may vary considerably from year to year for reasons other than changes in cancer occurrence. For instance, estimates are affected by changes in method, which are implemented regularly as modeling techniques improve and surveillance coverage becomes more complete. In addition, the model is sometimes oversensitive or undersensitive to abrupt or large changes in observed data. For these reasons, the projections are not an accurate measure of year-to-year changes in cancer occurrence and death. The data sources used for tracking cancer trends are age-standardized or age-specific cancer death rates from the NCHS and cancer incidence rates

from SEER and/or the National Program of Cancer Registries. Nevertheless, the American Cancer Society projections of new cancer cases and deaths provide a reasonably accurate estimate of the current cancer burden in the United States.

Errors in reporting race/ethnicity in medical records and on death certificates may result in underestimates of cancer incidence and mortality rates in nonwhite and nonblack populations. It is also important to note that cancer data in the United States are primarily reported for broad racial and ethnic groups that are not homogenous, masking important differences in the cancer burden within these groups.

Conclusions

Cancer death rates have been continuously declining for the past 2 decades. Overall, the risk of dying from cancer decreased by 20% between 1991 and 2010. Progress has been most rapid for middle-aged black men, among whom death rates have declined by approximately 50%. Despite this substantial progress, 5-year survival rates among blacks continue to lag behind whites by as much as 22 percentage points for uterine cancer, 21 percentage points for cancer of

the oral cavity, and 17 percentage points for urinary bladder cancer. Further progress can be accelerated by applying existing cancer control knowledge across all segments of the population, with an emphasis on those groups in the lowest socioeconomic bracket and other disadvantaged populations. ■

References

1. Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Mortality-All COD, Aggregated With State, Total US (1969-2010) <Katrina/Rita Population Adjustment>. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Cancer Statistics Branch; 2013. Released April 2013; underlying mortality data provided by National Center for Health Statistics 2013.
2. Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Incidence-SEER 9 Regs Public Use, Nov. 2011 Sub (1973-2010)-Linked to County Attributes-Total US, 1969-2011 Counties. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Cancer Statistics Branch; 2013.
3. Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Incidence-SEER 13 Regs Public Use, Nov. 2011 Sub (1992-2010)-Linked to County Attributes-Total US, 1969-2011 Counties. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Cancer Statistics Branch; 2013.
4. Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: Incidence-SEER 18 Regs Public Use, Nov. 2012 Sub (2000-2010)-Linked to County Attributes-Total US, 1969-2011 Counties. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Cancer Statistics Branch; 2013.
5. Howlader N, Noone AM, Krapcho M, et al, eds. SEER Cancer Statistics Review, 1975-2010. Bethesda, MD: National Cancer Institute; 2013.
6. Surveillance, Epidemiology, and End Results (SEER) Program. SEER*Stat Database: North American Association of Central Cancer Registries (NAACCR) Incidence-CiNA Analytic File, 1995-2010, for Expanded Races, custom file with county, ACS Facts & Figures projection project, North American Association of Central Cancer Registries. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Cancer Statistics Branch; 2013.
7. Copeland G, Lake A, Firth R, et al. Cancer in North America: 2006-2010. Vol 1. Combined Cancer Incidence for the United States, Canada and North America. Springfield, IL: North American Association of Central Cancer Registries Inc; 2013.
8. Copeland G, Lake A, Firth R, et al. Cancer in North America: 2006-2010. Vol 2. Registry-Specific Cancer Incidence in the United States and Canada. Springfield, IL: North American Association of Central Cancer Registries Inc; 2013.
9. Fritz A, Percy C, Jack A, et al. International Classification of Diseases for Oncology. 3rd ed. Geneva: World Health Organization; 2000.
10. Statistical Research and Applications Branch. DevCan: Probability of Developing or Dying of Cancer Software. Version 6.7.0. Bethesda, MD: Statistical Research and Applications Branch, National Cancer Institute; 2013.
11. Surveillance Research Program, National Cancer Institute. SEER*Stat Software. Version 8.1.2. Bethesda, MD: National Cancer Institute; 2013.
12. Clegg LX, Feuer EJ, Midthune DN, Fay MP, Hankey BF. Impact of reporting delay and reporting error on cancer incidence rates and trends. *J Natl Cancer Inst.* 2002;94:1537-1545.
13. Pickle LW, Hao Y, Jemal A, et al. A new method of estimating United States and state-level cancer incidence counts for the current calendar year. *CA Cancer J Clin.* 2007;57:30-42.
14. Zhu L, Pickle LW, Ghosh K, et al. Predicting US- and state-level cancer counts for the current calendar year: Part II: evaluation of spatiotemporal projection methods for incidence. *Cancer.* 2012;118:1100-1109.
15. Joinpoint Regression Program, Version 4.0.4. Bethesda, MD: Statistical Research and Applications Branch, National Cancer Institute; 2013.
16. Chen HS, Portier K, Ghosh K, et al. Predicting US- and state-level cancer counts for the current calendar year: Part I: evaluation of temporal projection methods for mortality. *Cancer.* 2012;118:1091-1099.
17. Kim HJ, Fay MP, Feuer EJ, Midthune DN. Permutation tests for joinpoint regression with applications to cancer rates. *Stat Med.* 2000;19:335-351.
18. Edwards BK, Ward E, Kohler BA, et al. Annual report to the nation on the status of cancer, 1975-2006, featuring colorectal cancer trends and impact of interventions (risk factors, screening, and treatment) to reduce future rates. *Cancer.* 2010;116:544-573.
19. Cress RD, Morris C, Ellison GL, Goodman MT. Secular changes in colorectal cancer incidence by subsite, stage at diagnosis, and race/ethnicity, 1992-2001. *Cancer.* 2006;107(suppl 5):1142-1152.
20. Siegel RL, Ward EM, Jemal A. Trends in colorectal cancer incidence rates in the United States by tumor location and stage, 1992-2008. *Cancer Epidemiol Biomarkers Prev.* 2012;21:411-416.
21. Weiss W. Cigarette smoking and lung cancer trends. A light at the end of the tunnel? *Chest.* 1997;111:1414-1416.
22. Simard EP, Ward EM, Siegel R, Jemal A. Cancers with increasing incidence trends in the United States: 1999 through 2008. *CA Cancer J Clin.* 2012;62:118-128.
23. Jemal A, Thun MJ, Ries LA, et al. Annual report to the nation on the status of cancer, 1975-2005, featuring trends in lung cancer, tobacco use, and tobacco control. *J Natl Cancer Inst.* 2008;100:1672-1694.
24. Berry DA, Cronin KA, Plevritis SK, et al; Cancer Intervention and Surveillance Modeling Network (CISNET) Collaborators. Effect of screening and adjuvant therapy on mortality from breast cancer. *N Engl J Med.* 2005;353:1784-1792.
25. Etzioni R, Tsodikov A, Mariotto A, et al. Quantifying the role of PSA screening in the US prostate cancer mortality decline. *Cancer Causes Control.* 2008;19:175-181.
26. Potosky AL, Miller BA, Albertsen PC, Kramer BS. The role of increasing detection in the rising incidence of prostate cancer. *JAMA.* 1995;273:548-552.
27. Ward E, Jemal A, Cokkinides V, et al. Cancer disparities by race/ethnicity and socioeconomic status. *CA Cancer J Clin.* 2004;54:78-93.
28. Ghafoor A, Jemal A, Ward E, Cokkinides V, Smith R, Thun M. Trends in breast cancer by race and ethnicity. *CA Cancer J Clin.* 2003;53:342-355.
29. Menashe I, Anderson WF, Jatoi I, Rosenberg PS. Underlying causes of the black-white racial disparity in breast cancer mortality: a population-based analysis. *J Natl Cancer Inst.* 2009;101:993-1000.
30. Press R, Carrasquillo O, Sciacca RR, Giardina EG. Racial/ethnic disparities in time to follow-up after an abnormal mammogram. *J Womens Health (Larchmt).* 2008;17:923-930.
31. Tammemagi CM, Nerenz D, Neslund-Dudas C, Feldkamp C, Nathanson D. Comorbidity and survival disparities among black and white patients with breast cancer. *JAMA.* 2005;294:1765-1772.
32. Parkin DM. The global health burden of infection-associated cancers in the year 2002. *Int J Cancer.* 2006;118:3030-3044.
33. Espey DK, Wu XC, Swan J, et al. Annual report to the nation on the status of cancer, 1975-2004, featuring cancer in American Indians and Alaska Natives. *Cancer.* 2007;110:2119-2152.
34. Bach PB, Schrag D, Brawley OW, Galaznik A, Yakren S, Begg CB. Survival of blacks and whites after a cancer diagnosis. *JAMA.* 2002;287:2106-2113.
35. Byers TE, Wolf HJ, Bauer KR, et al; Patterns of Care Study Group. The impact of socioeconomic status on survival after cancer in the United States: findings from the National Program of Cancer Registries Patterns of Care Study. *Cancer.* 2008;113:582-591.
36. Schwartz KL, Crossley-May H, Vigneau FD, Brown K, Banerjee M. Race, socioeconomic status and stage at diagnosis for five common malignancies. *Cancer Causes Control.* 2003;14:761-766.
37. Gross CP, Smith BD, Wolf E, Andersen M. Racial disparities in cancer therapy: did the gap narrow between 1992 and 2002? *Cancer.* 2008;112:900-908.